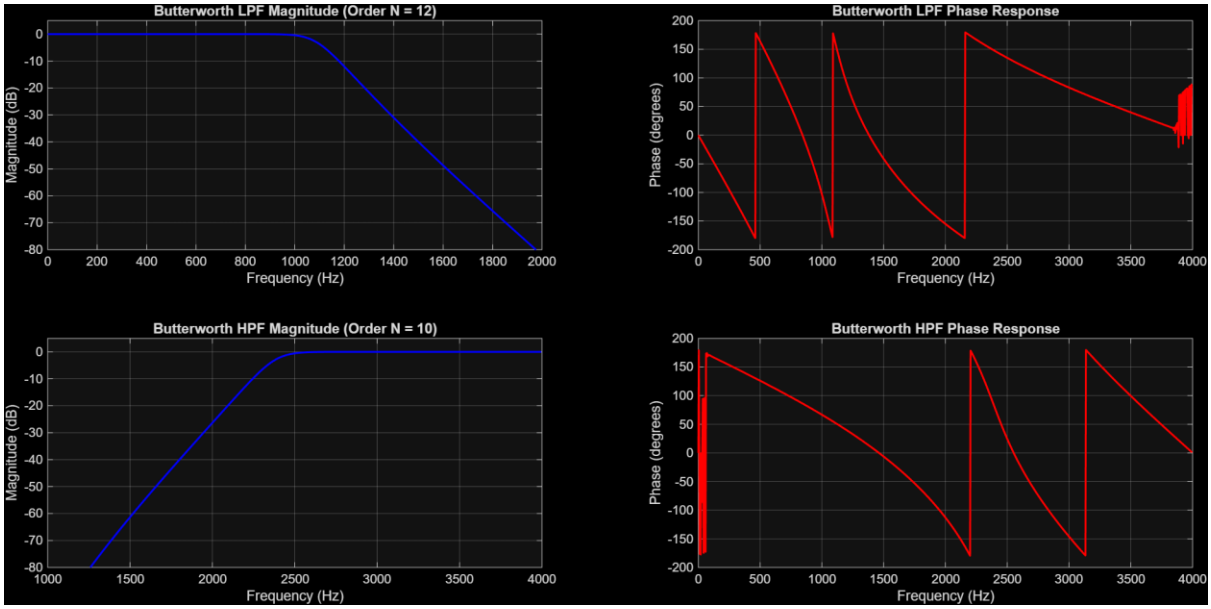


Experiment 1 : Design, Realization and Performance Analysis of Butterworth IIR Low-Pass and High-Pass Digital Filters Using MATLAB

Command Window

New to MATLAB? See resources for Getting Started.

```
=== EXPERIMENT 1: BUTTERWORTH IIR FILTERS ===

--- LOW-PASS FILTER ---
Calculated Filter Order (N_lp): 12
3-dB Cutoff Frequency (Wn_lp): 1087.8354 Hz
Numerator Coefficients (b_lp):
  Columns 1 through 9

    0.0000    0.0000    0.0002    0.0006    0.0013    0.0021    0.0024    0.0021    0.0013

  Columns 10 through 13

    0.0006    0.0002    0.0000    0.0000

Denominator Coefficients (a_lp):
  Columns 1 through 9

    1.0000   -5.4622   14.6771  -25.1616   30.3497  -26.9537   17.9862   -9.0534    3.4018

  Columns 10 through 13

   -0.9283    0.1743   -0.0202    0.0011

--- HIGH-PASS FILTER ---
Calculated Filter Order (N_hp): 10
3-dB Cutoff Frequency (Wn_hp): 2379.7556 Hz
Numerator Coefficients (b_hp):
  Columns 1 through 9

    0.0006   -0.0055    0.0248   -0.0662    0.1159   -0.1390    0.1159   -0.0662    0.0248

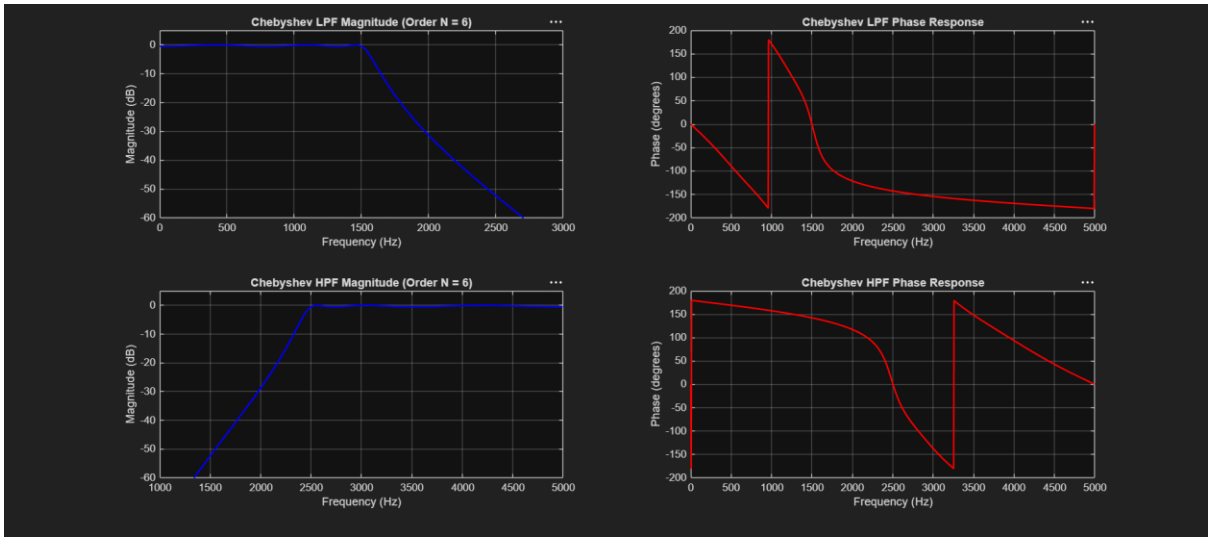
  Columns 10 through 11

   -0.0055    0.0006
```

Result

Experiment 2 : Design and Performance Analysis of Chebyshev Type-I IIR Low-Pass and High-Pass Filters Using MATLAB

Output:



Result:

```
Command Window
New to MATLAB? See resources for Getting Started.

=== EXPERIMENT 2: CHEBYSHEV TYPE-I IIR FILTERS ===

--- CHEBYSHEV LOW-PASS FILTER ---
Calculated Filter Order (N_lp): 6
Passband Edge Cutoff Frequency (Wn_lp): 1500.0000 Hz
Numerator Coefficients (b_lp):
    0.0006    0.0038    0.0095    0.0127    0.0095    0.0038    0.0006

Denominator Coefficients (a_lp):
    1.0000   -3.7650    6.8198   -7.3274    4.8717   -1.8927    0.3368

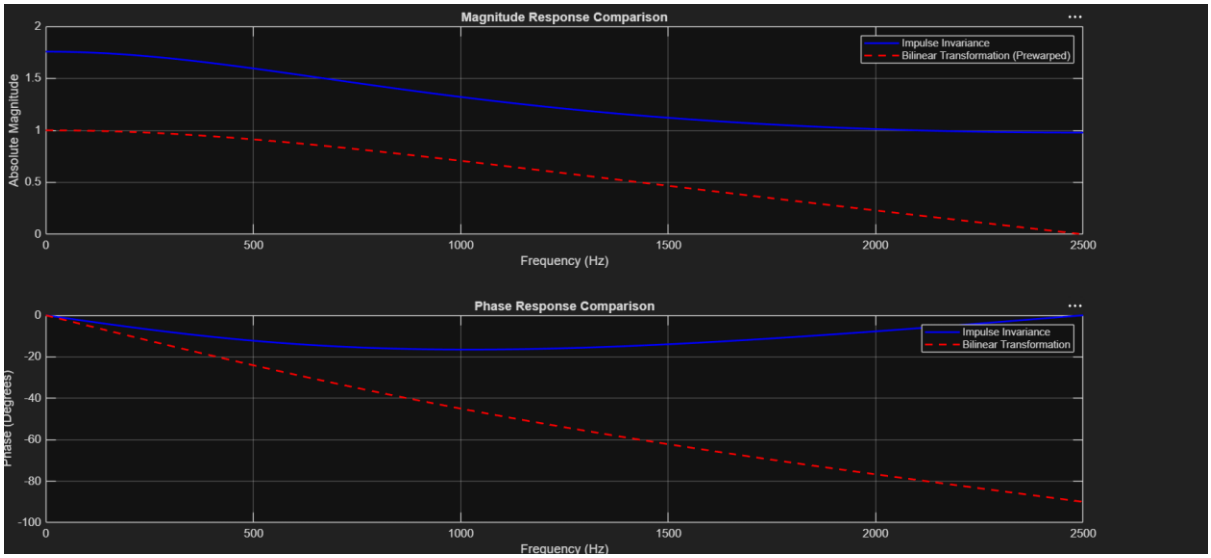
--- CHEBYSHEV HIGH-PASS FILTER ---
Calculated Filter Order (N_hp): 6
Passband Edge Cutoff Frequency (Wn_hp): 2500.0000 Hz
Numerator Coefficients (b_hp):
    0.0117   -0.0704    0.1761   -0.2348    0.1761   -0.0704    0.0117

Denominator Coefficients (a_hp):
    1.0000    1.3568    2.1347    1.8511    1.2978    0.5937    0.1650

>>
```

Experiment 3 : Design of IIR Filters Using Impulse Invariance and Bilinear Transformation Methods Using MATLAB

Output:



Result:

```
Command Window
New to MATLAB? See resources for Getting Started.
==========================================
EXPERIMENT 3: IMPULSE INVARIANCE vs BILINEAR
==========================================

Sampling Frequency: 5000.00 Hz
Cutoff Frequency: 1000.00 Hz
Analog Cutoff Frequency: 6283.19 rad/s
Prewarped Analog Frequency: 7265.43 rad/s

--- IMPULSE INVARIANCE COEFFICIENTS ---
Numerator:
    1.2566

Denominator:
    1.0000   -0.2846

--- BILINEAR TRANSFORMATION COEFFICIENTS ---
Numerator:
    0.4208    0.4208

Denominator:
    1.0000   -0.1584


Simulation completed successfully!
>>
```

Experiment 4: Design and Comparative Analysis of Butterworth, Chebyshev and Elliptic IIR Filters Using MATLAB

Output:

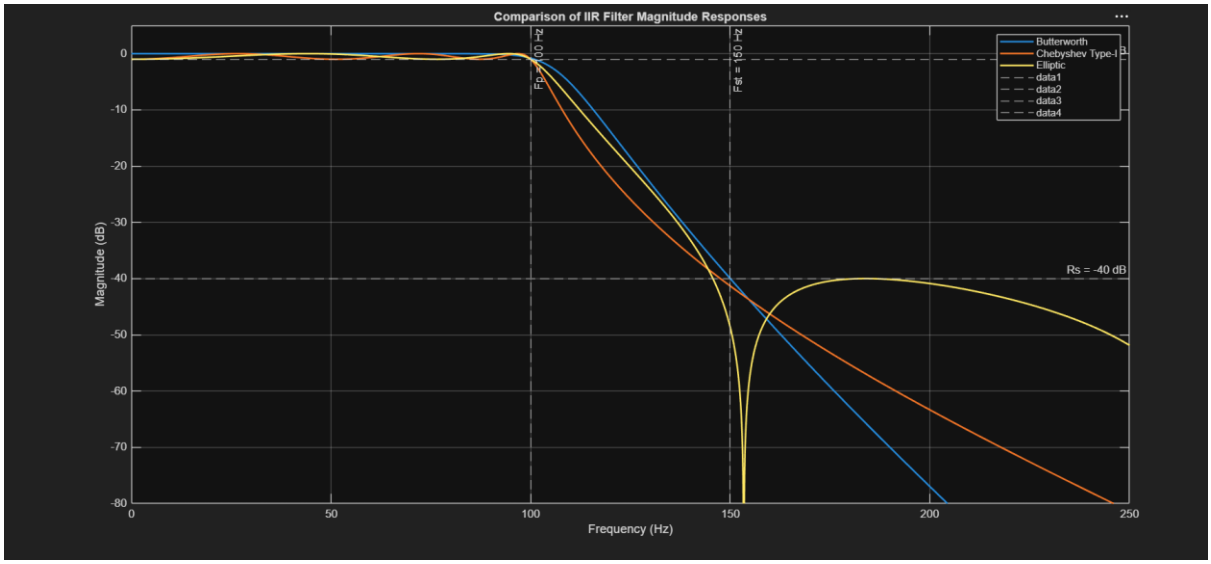


Fig.4.1

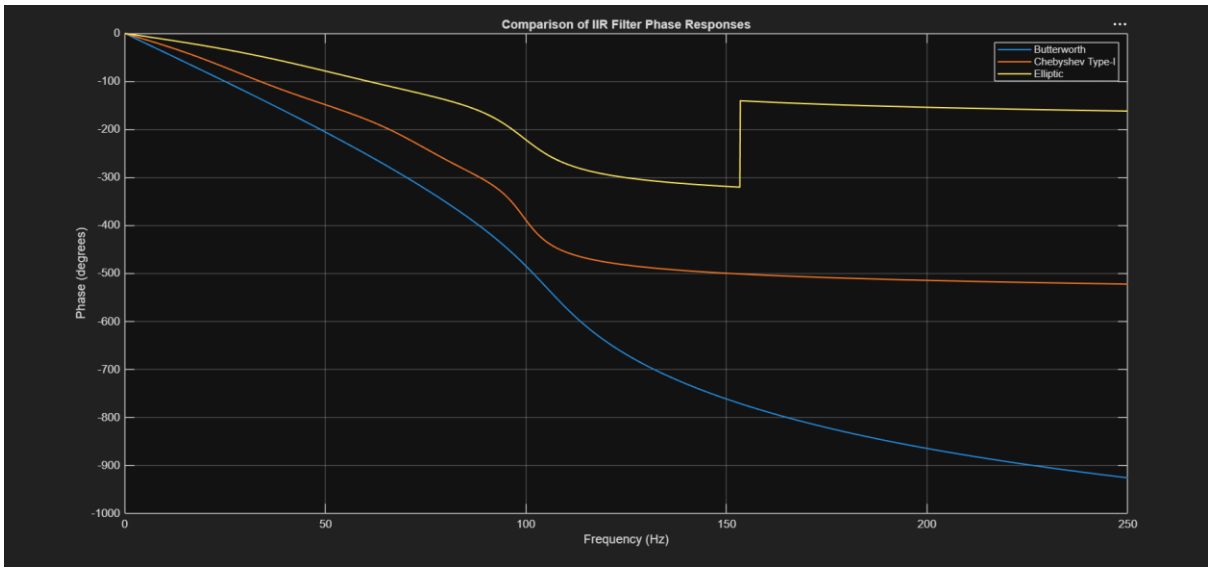


Fig.4.2

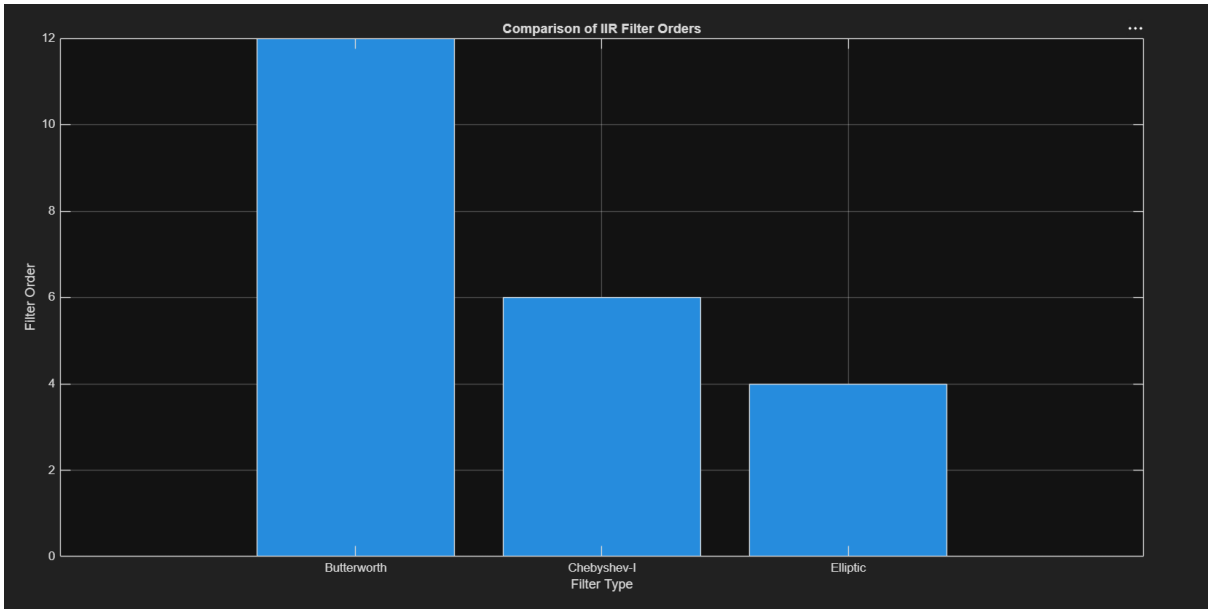


Fig.4.3

Experiment 5: Realization of IIR Filters Using Direct Form-I, Direct Form-II, Cascade and Parallel Structures Using MATLAB

Output:

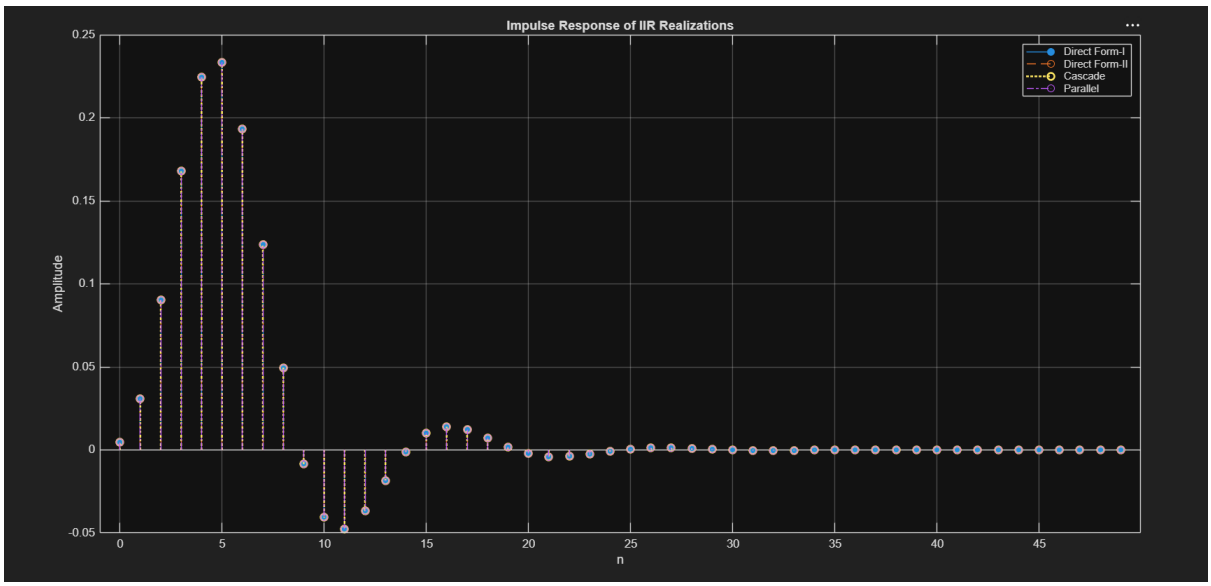


Fig.5.1

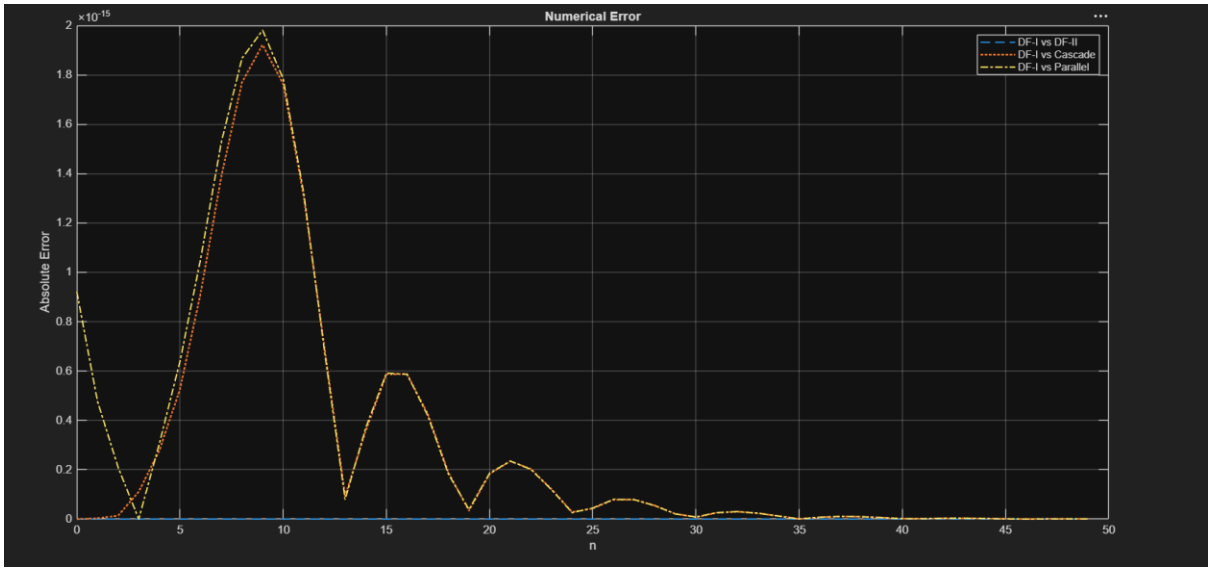

Numerical Error
×10^-15
Absolute Error
n
DF-I vs DF-II
DF-I vs Cascade
DF-I vs Parallel

Fig.5.2

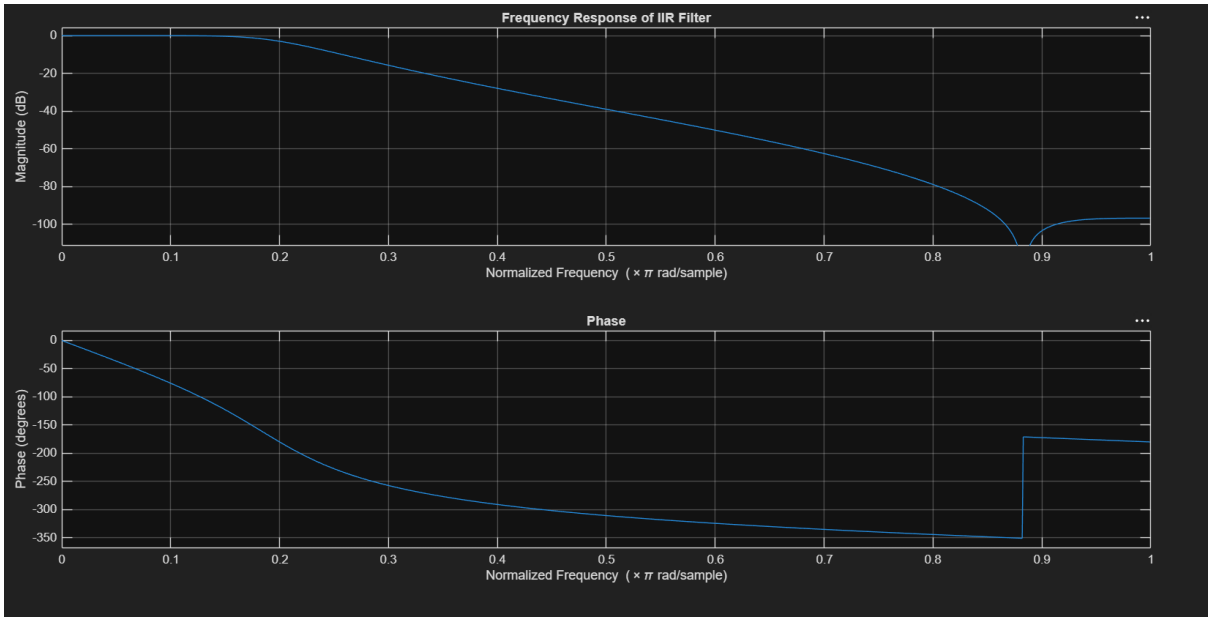

Frequency Response of IIR Filter
Magnitude (dB)
Normalized Frequency (× π rad/sample)
Phase
Phase (degrees)
Normalized Frequency (× π rad/sample)

Fig.5.3

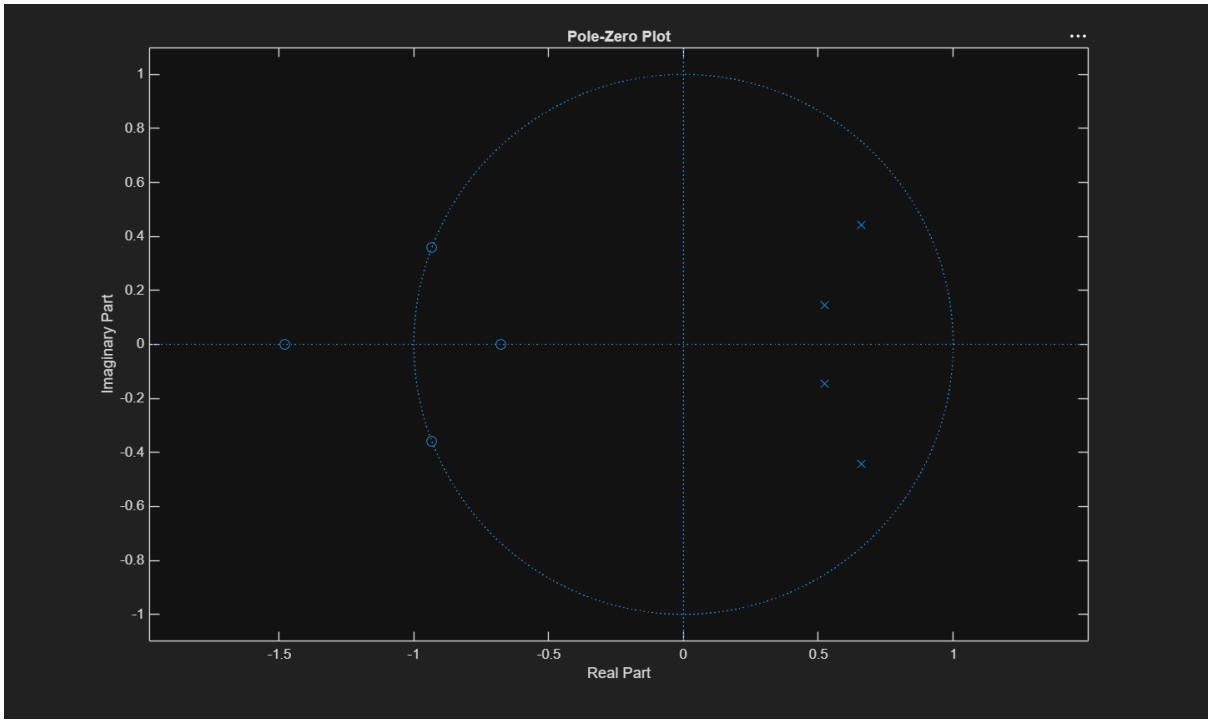
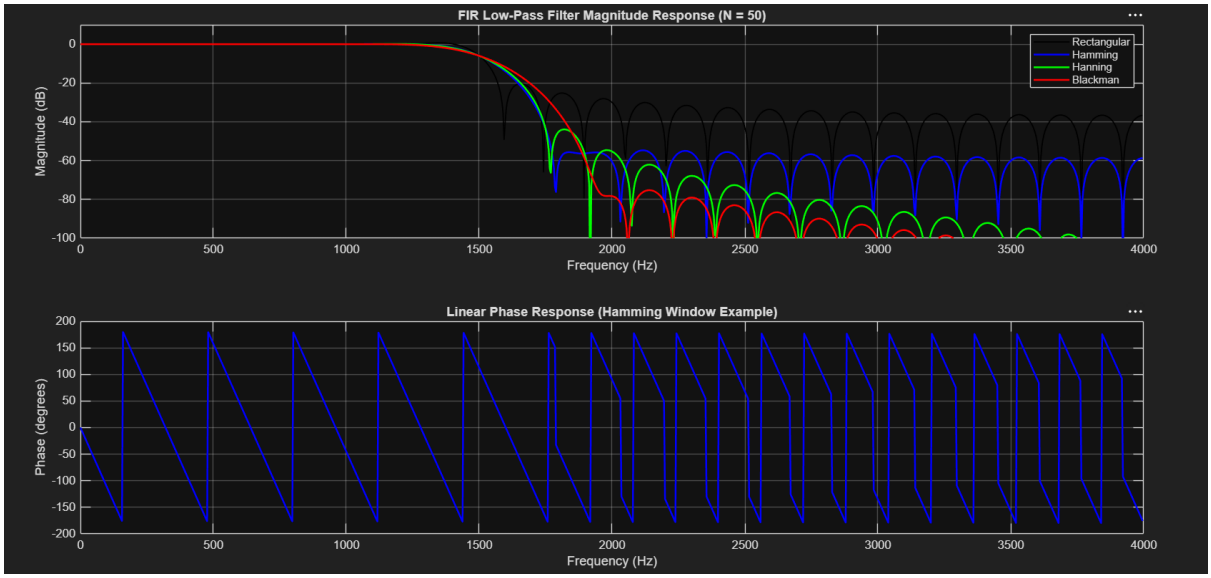

Pole-Zero Plot
Imaginary Part
Real Part
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1
-1.5
-1
-0.5
0
0.5
1

Fig.5.4

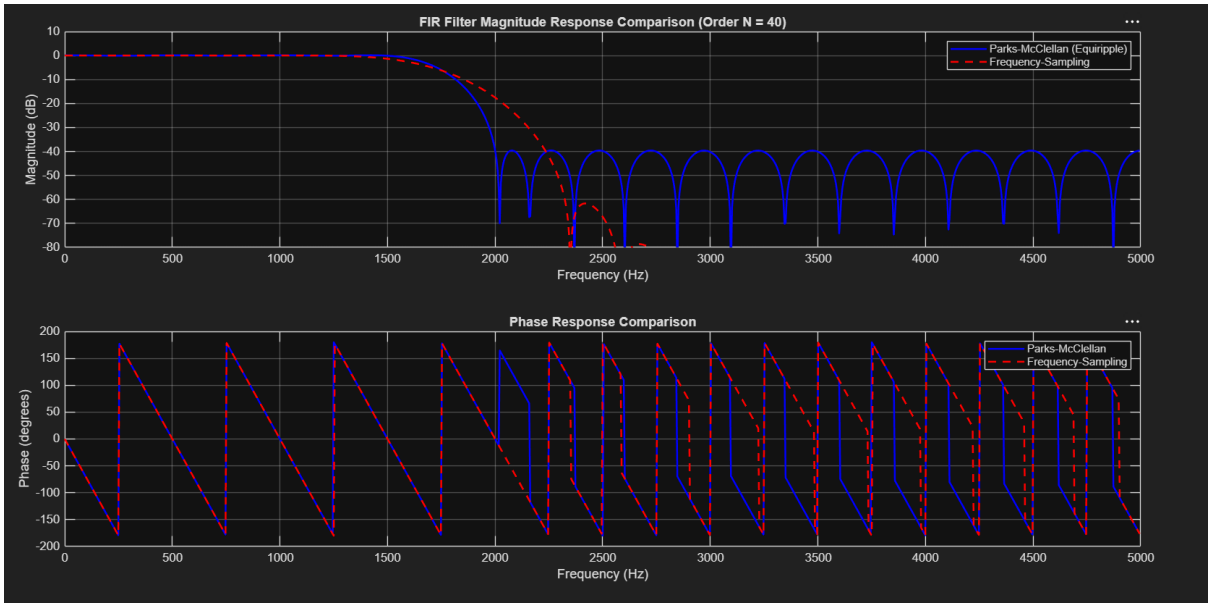
Experiment 6: Design and Analysis of FIR Digital Filters Using Fourier Series and Windowing Techniques Using MATLAB

Output:



Experiment 7: Design and Analysis of FIR Filters Using Parks–McClellan and Frequency-Sampling Techniques Using MATLAB

Output:



Experiment 8: Realization of FIR Filters Using Direct, Cascade and Linear-Phase Structures Using MATLAB

Output:

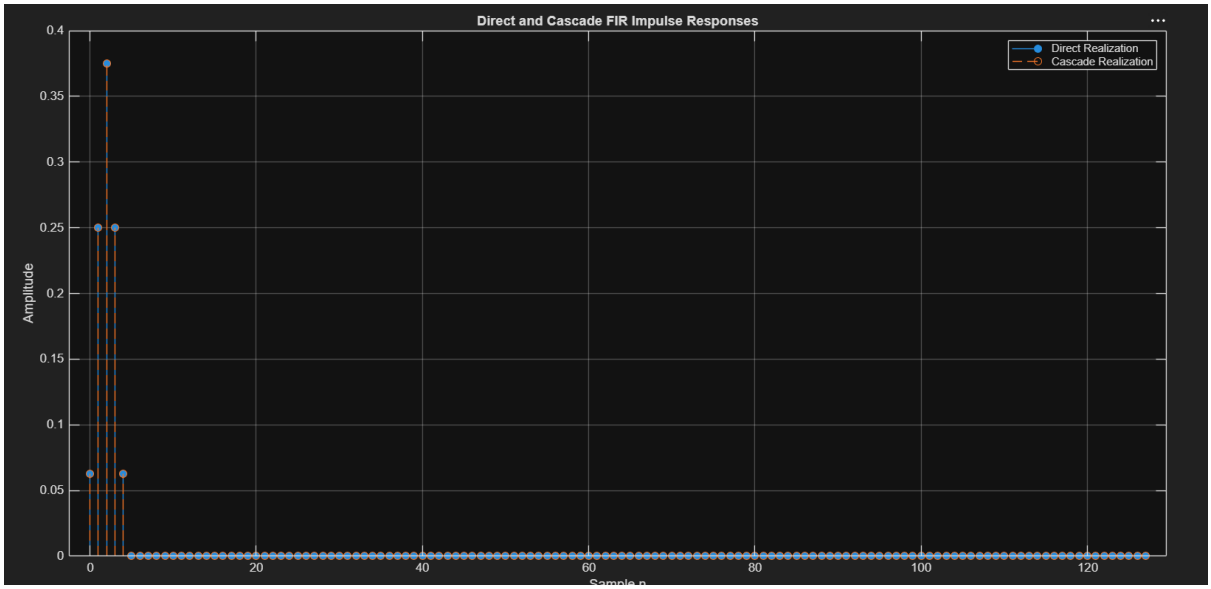


Fig.8.1

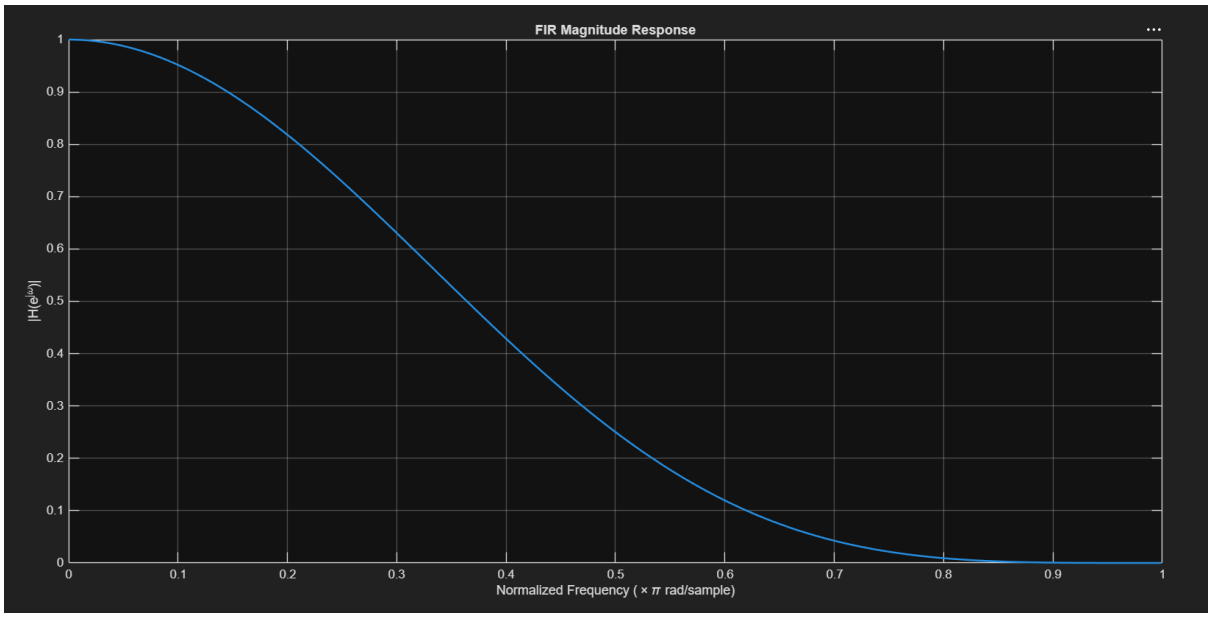


Fig.8.2

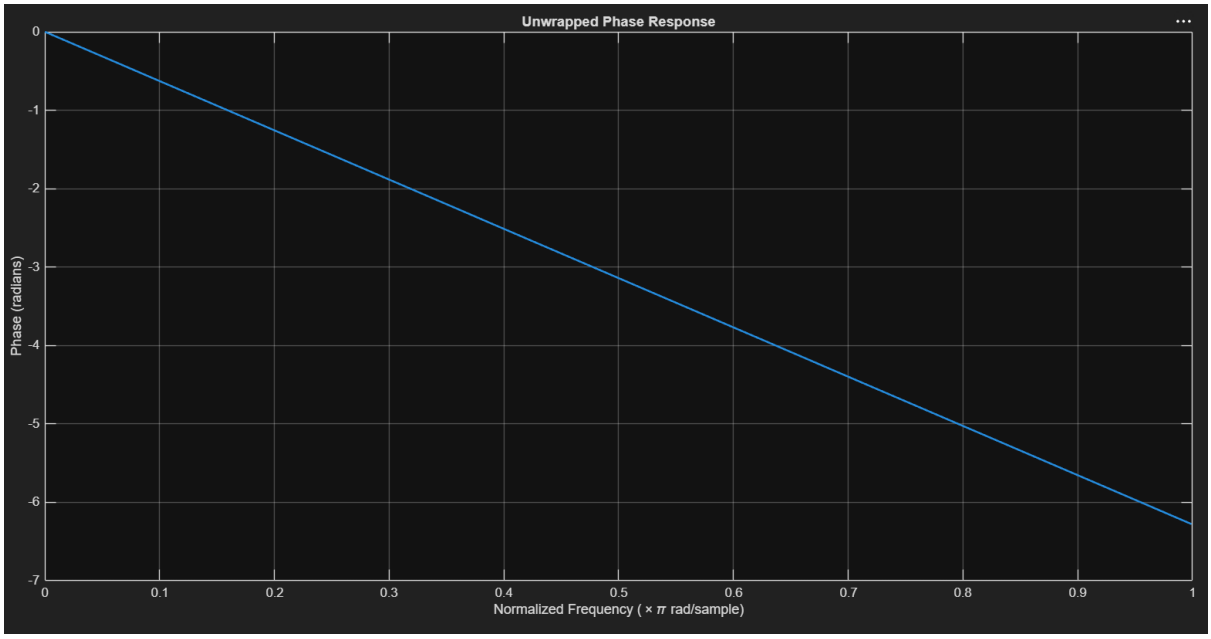

Unwrapped Phase Response
Phase (radians)
0
-1
-2
-3
-4
-5
-6
-7
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Normalized Frequency (× π rad/sample)

Fig.8.3

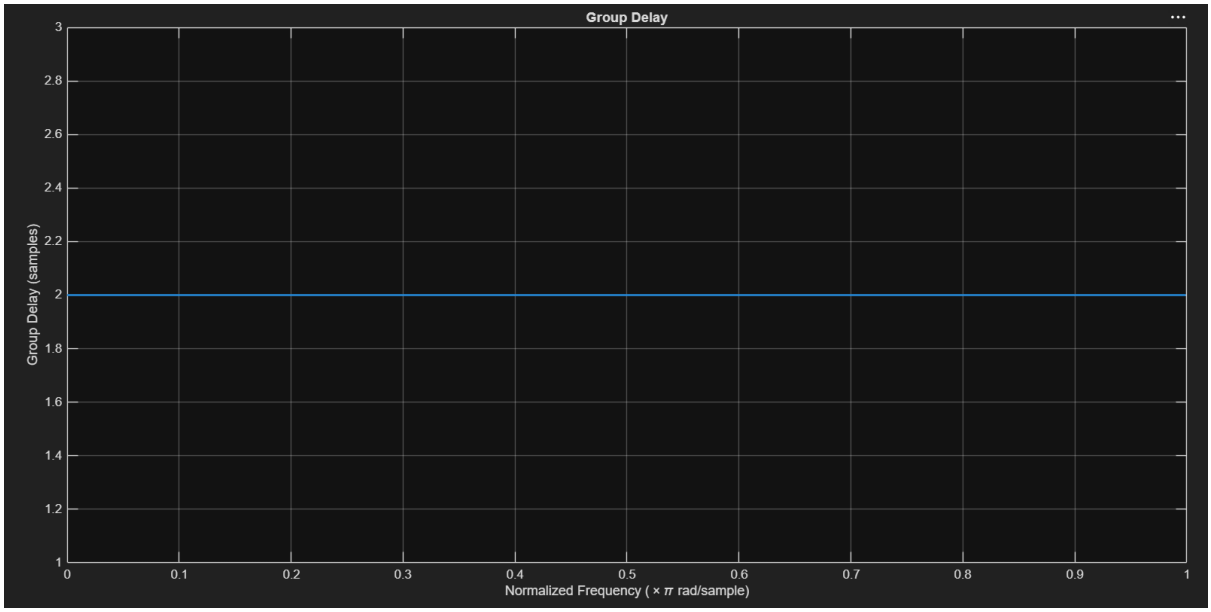

Group Delay
Group Delay (samples)
3
2.8
2.6
2.4
2.2
2
1.8
1.6
1.4
1.2
1
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Normalized Frequency (× π rad/sample)

Fig.8.4

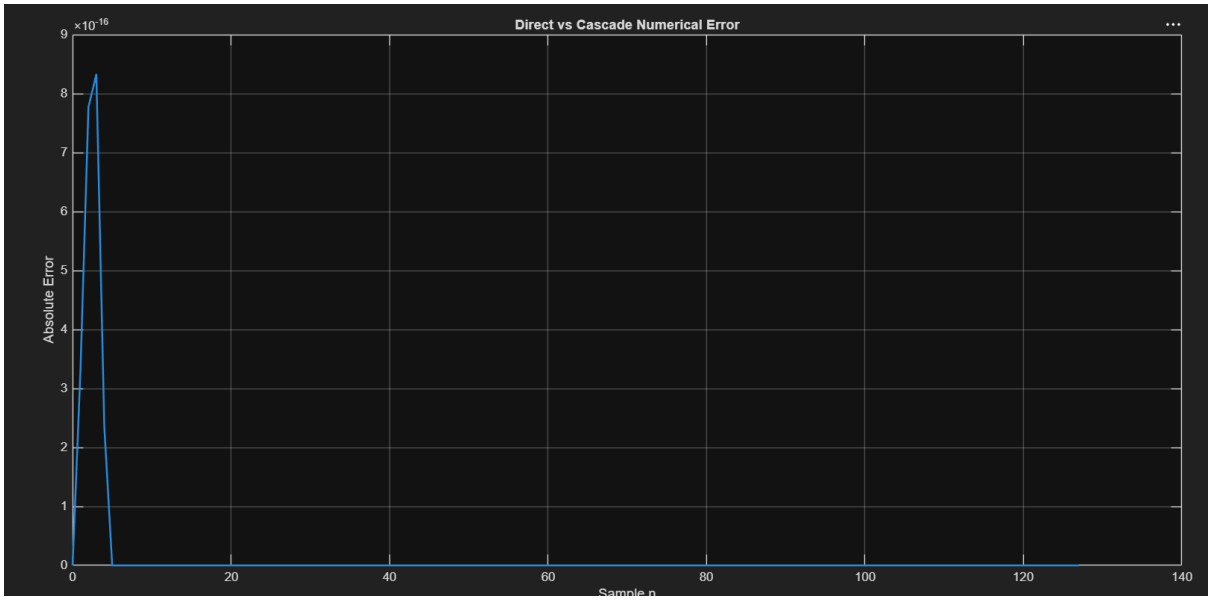

Direct vs Cascade Numerical Error
×10^-16
Absolute Error
Sample n
0
1
2
3
4
5
6
7
8
9
0
20
40
60
80
100
120
140

Fig.8.5

Experiment 9: Uniform Quantization and Finite Word-Length Effects in Digital Signal Processing Using MATLAB

Output:

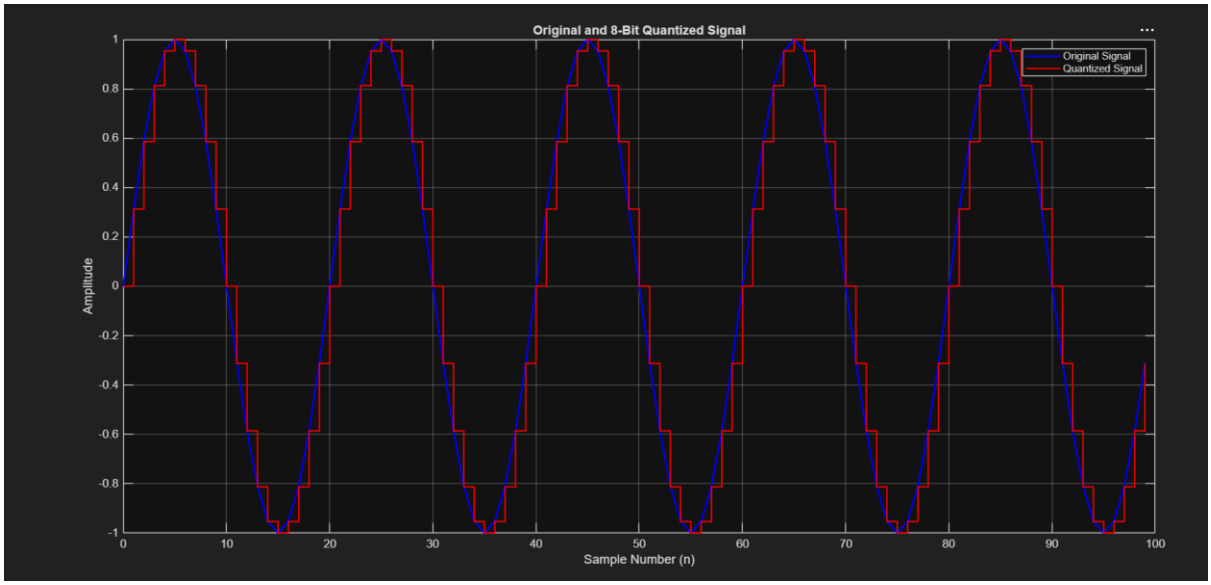


Fig.9.1

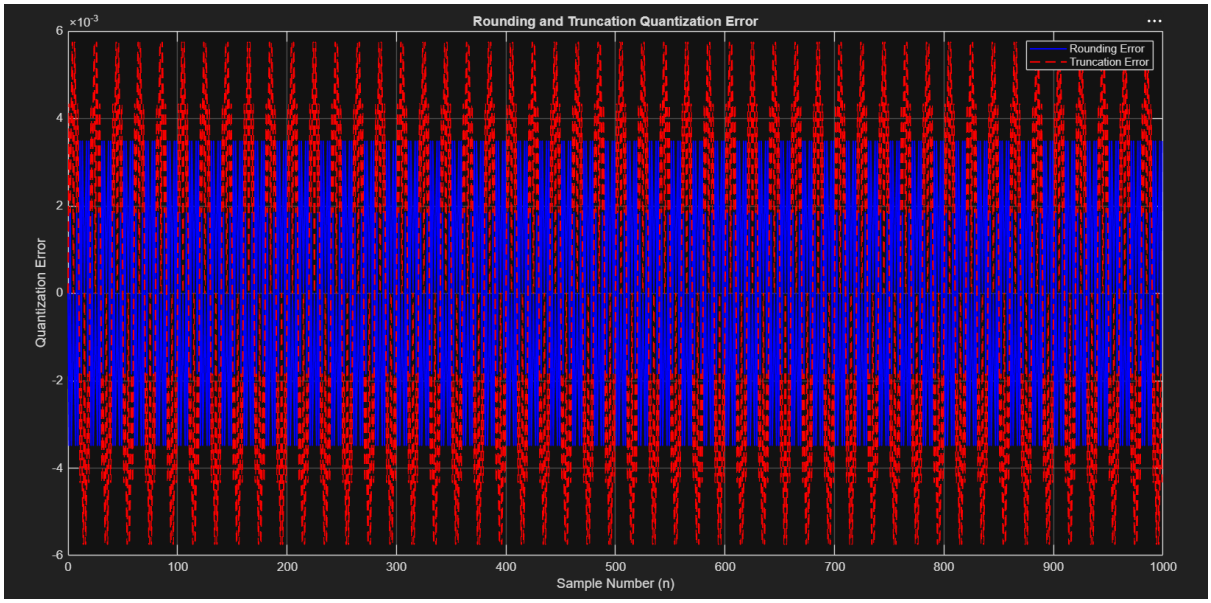


Fig.9.2

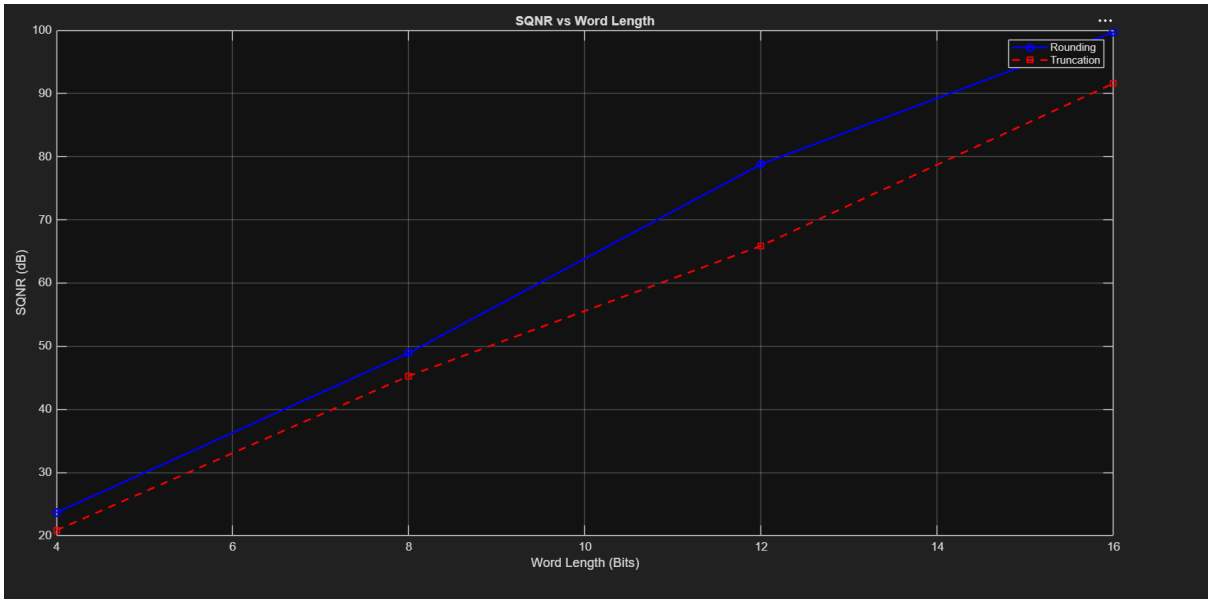

SQNR vs Word Length
Rounding
Truncation
SQNR (dB)
Word Length (Bits)
100
90
80
70
60
50
40
30
20
4
6
8
10
12
14
16

Fig.9.3

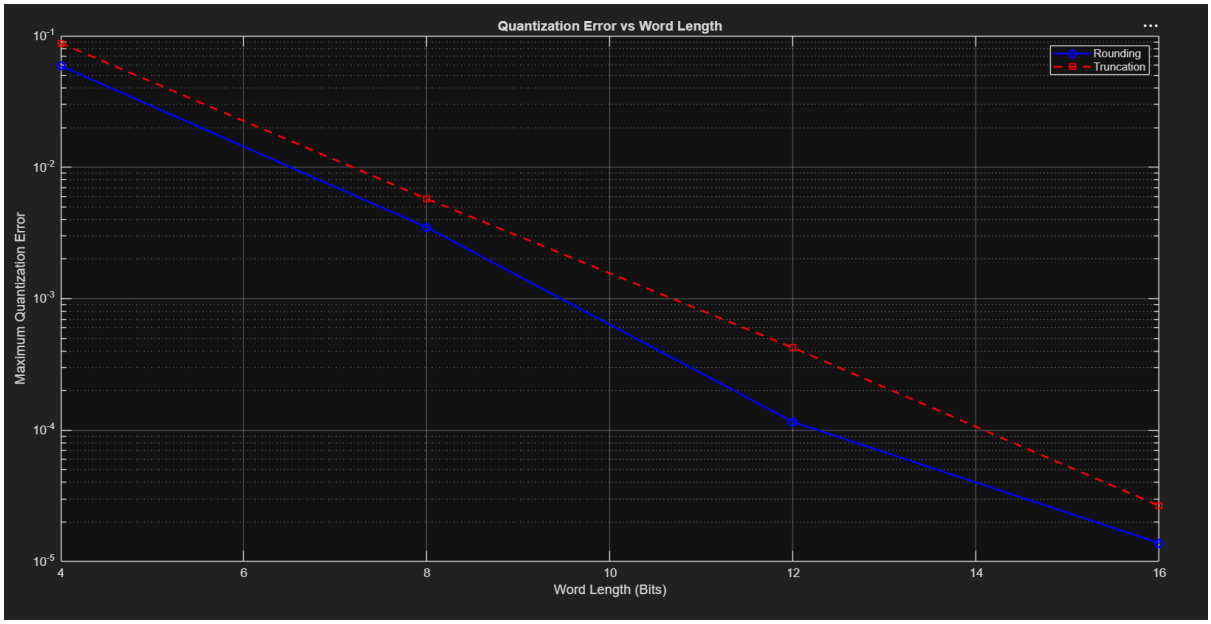

Quantization Error vs Word Length
Rounding
Truncation
Maximum Quantization Error
Word Length (Bits)
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
4
6
8
10
12
14
16

Fig.9.4

Experiment 10: Investigation of Coefficient and Product Quantization Noise and Limit-Cycle Oscillations Using MATLAB

Output:

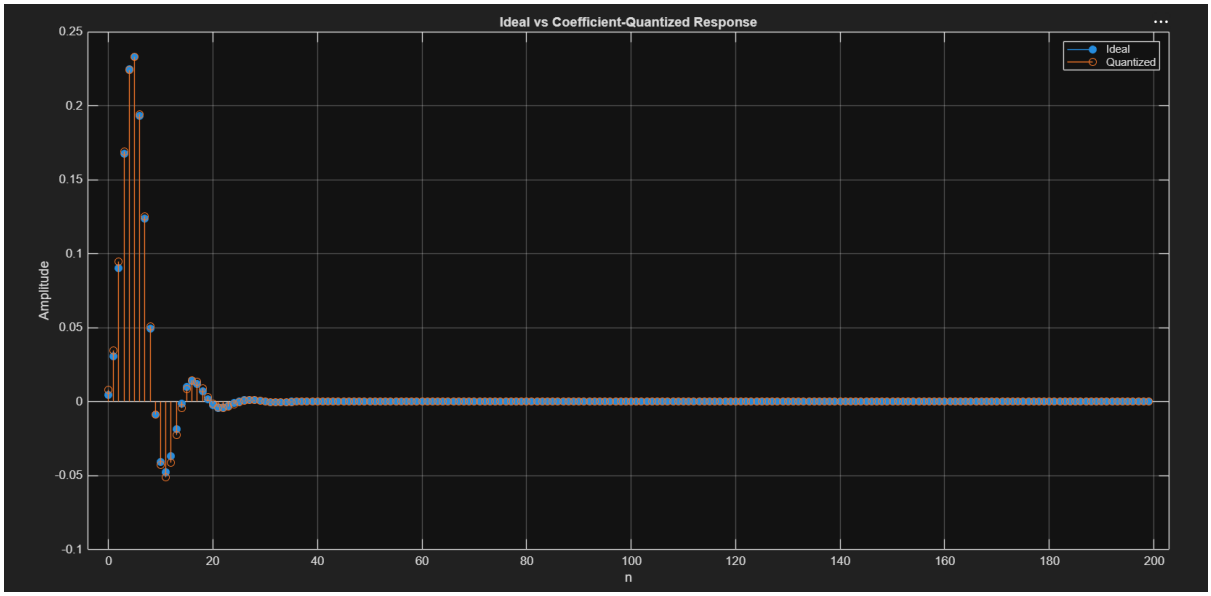


Fig.10.1

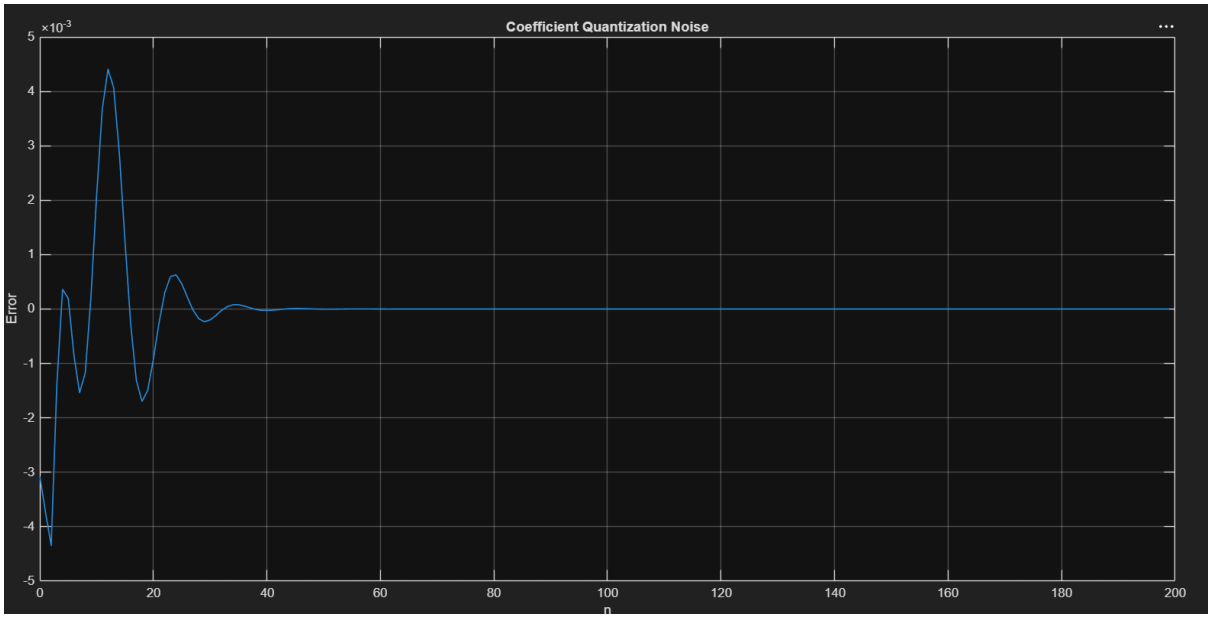


Fig10.2

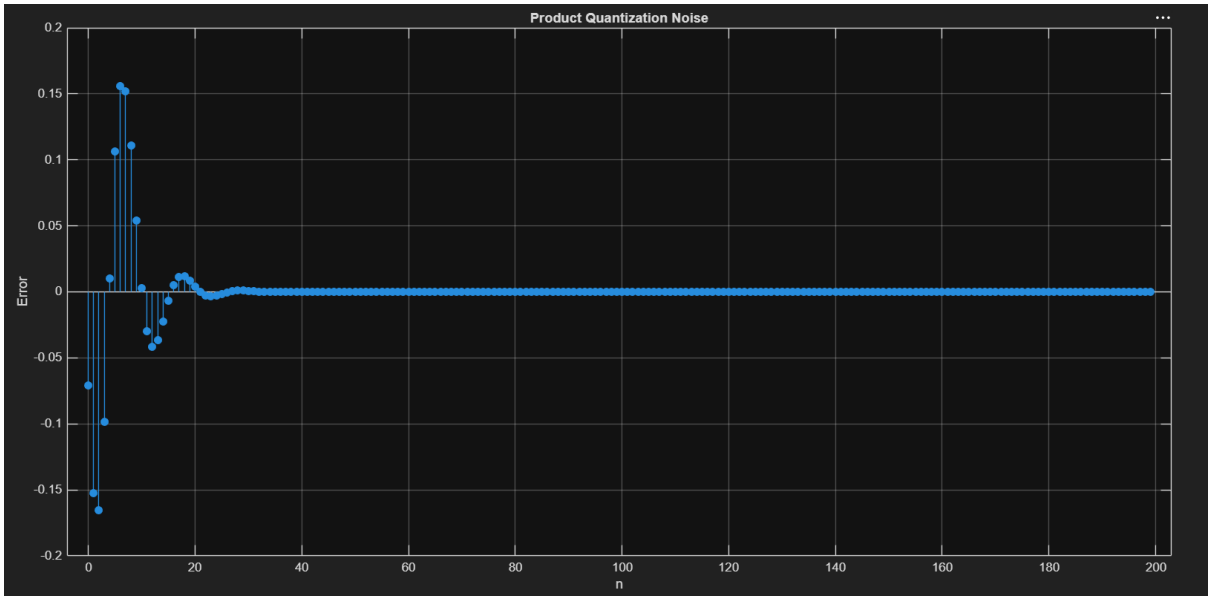


Fig.10.3

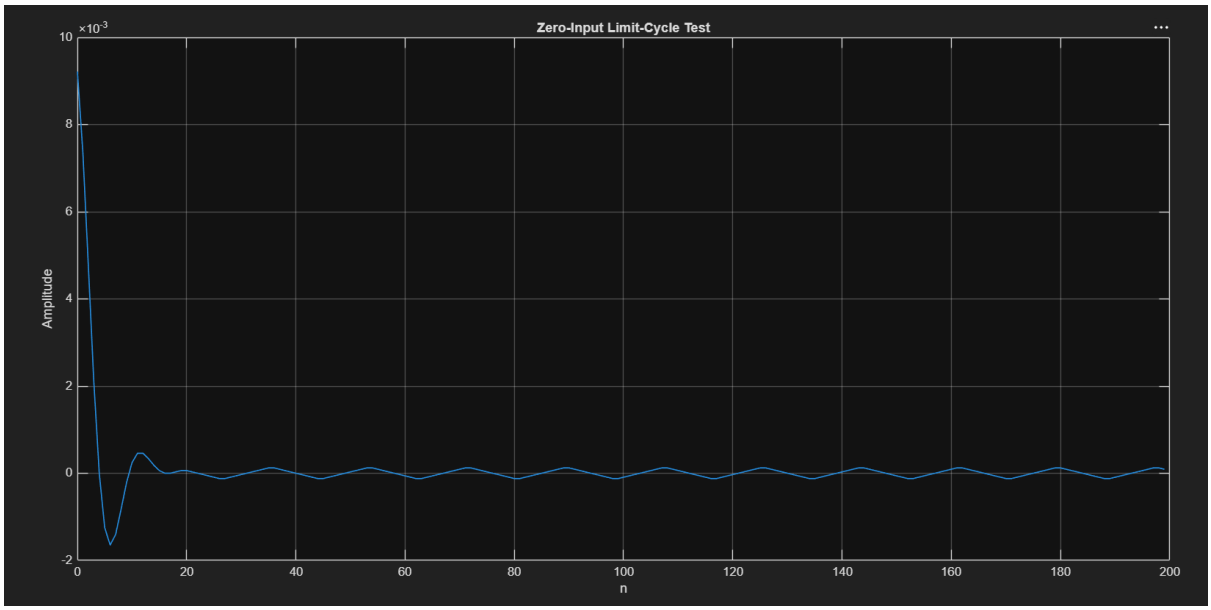


Fig.10.4

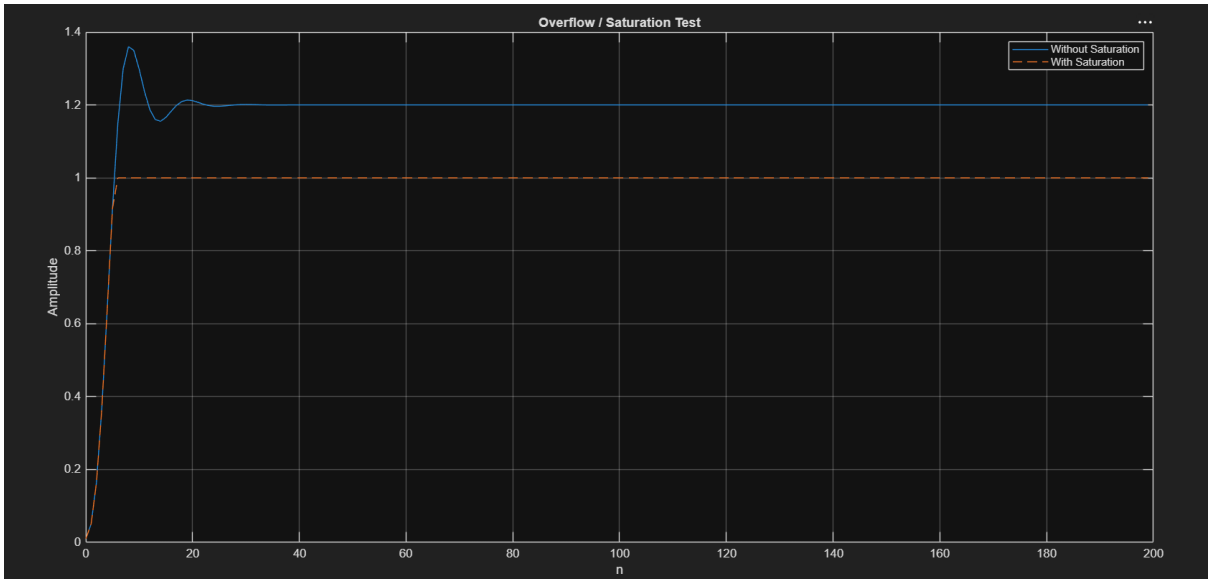


Fig.10.5

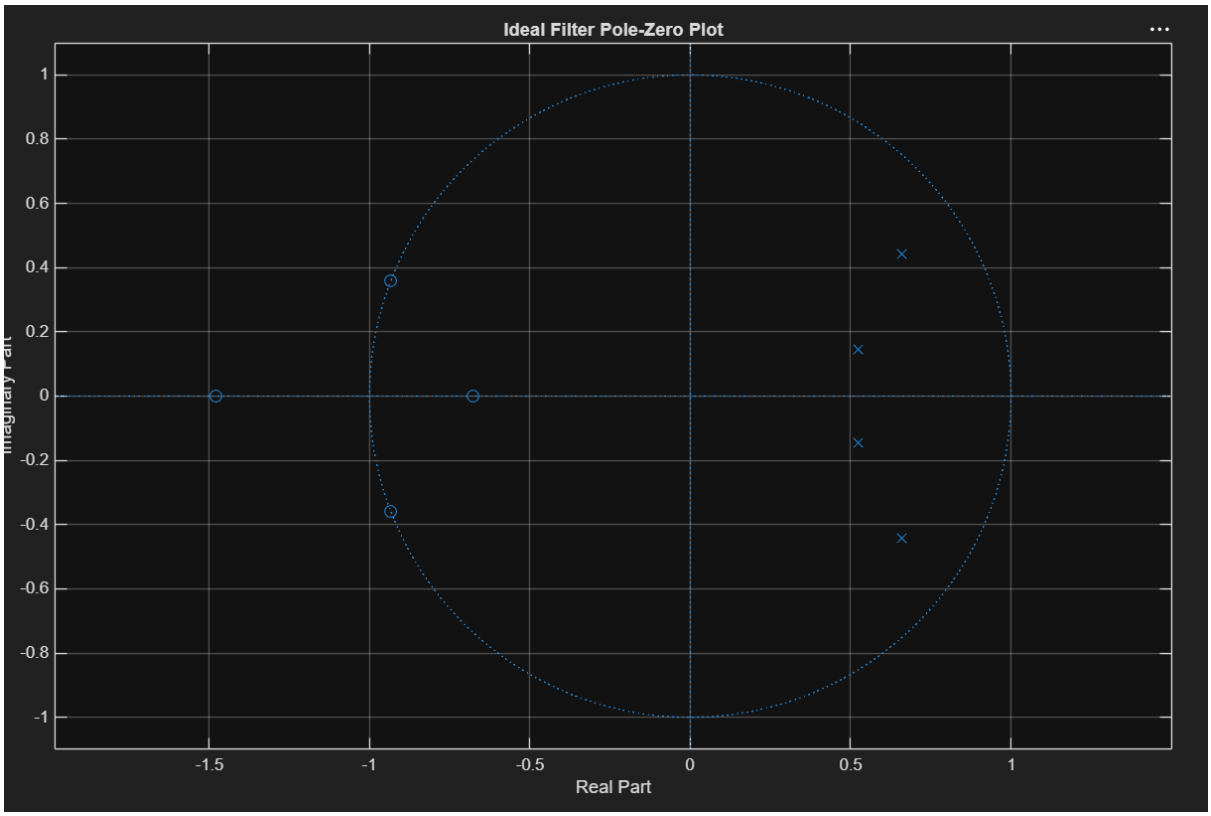


Fig.10.6

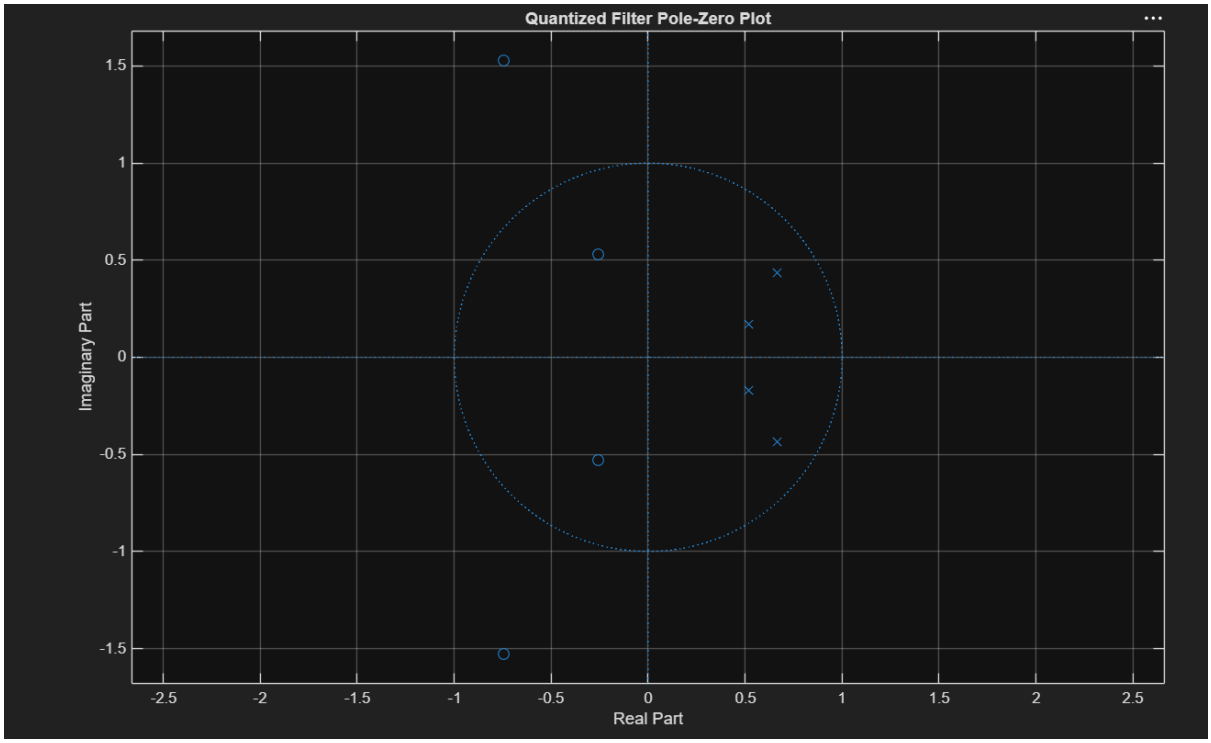

Quantized Filter Pole-Zero Plot
Imaginary Part
Real Part
1.5
1
0.5
0
-0.5
-1
-1.5
-2.5
-2
-1.5
-1
-0.5
0
0.5
1
1.5
2
2.5

Fig.10.7

Experiment 11: Investigation of Overflow Effects and Signal Scaling in Digital Filters Using MATLAB

Output:

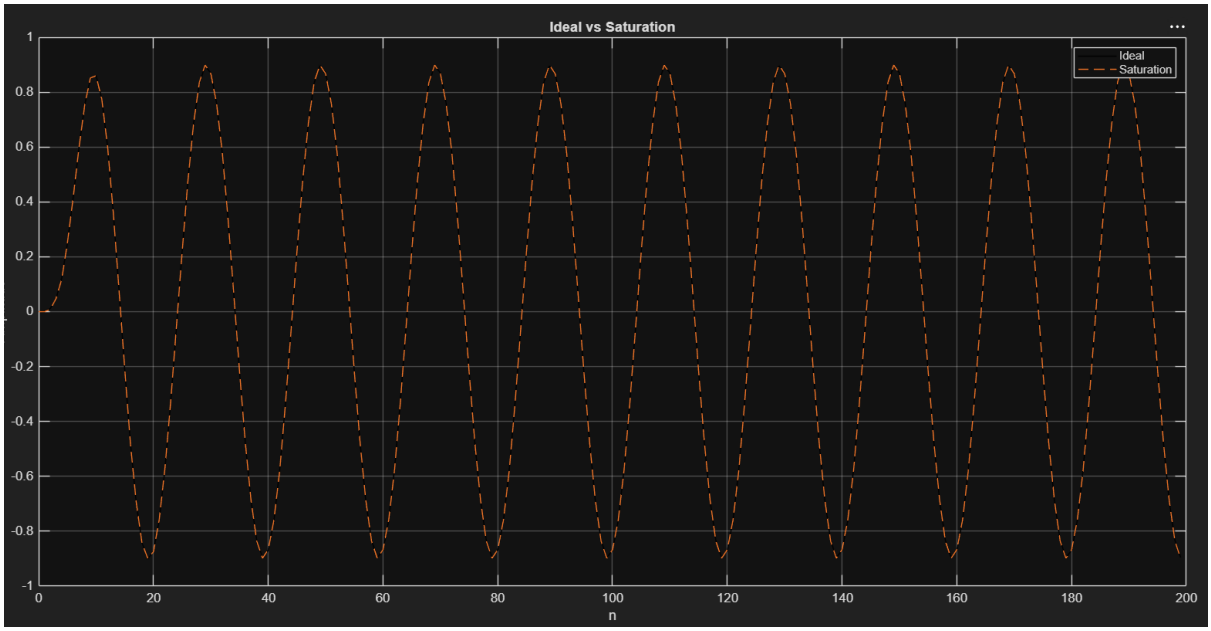


Fig.11.1

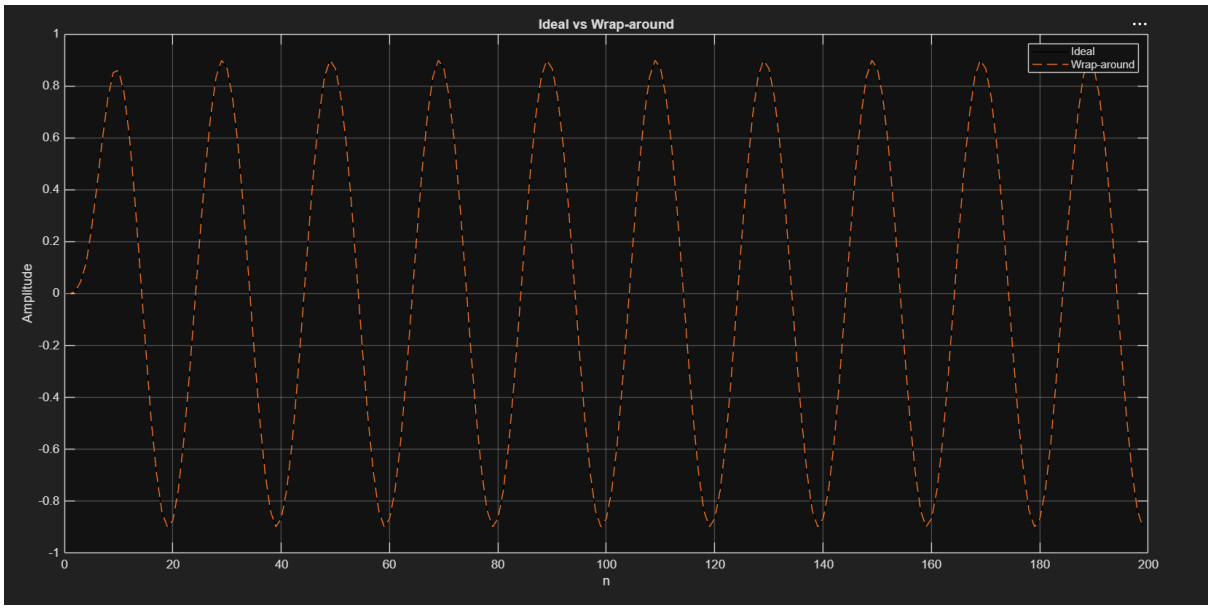


Fig.11.2

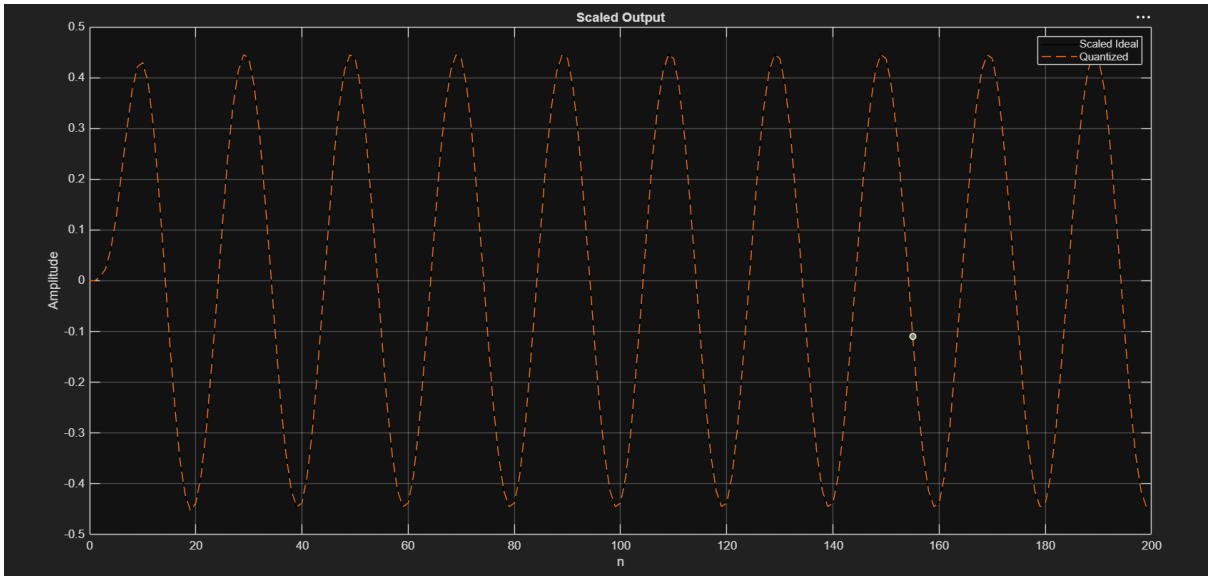


Fig.11.3

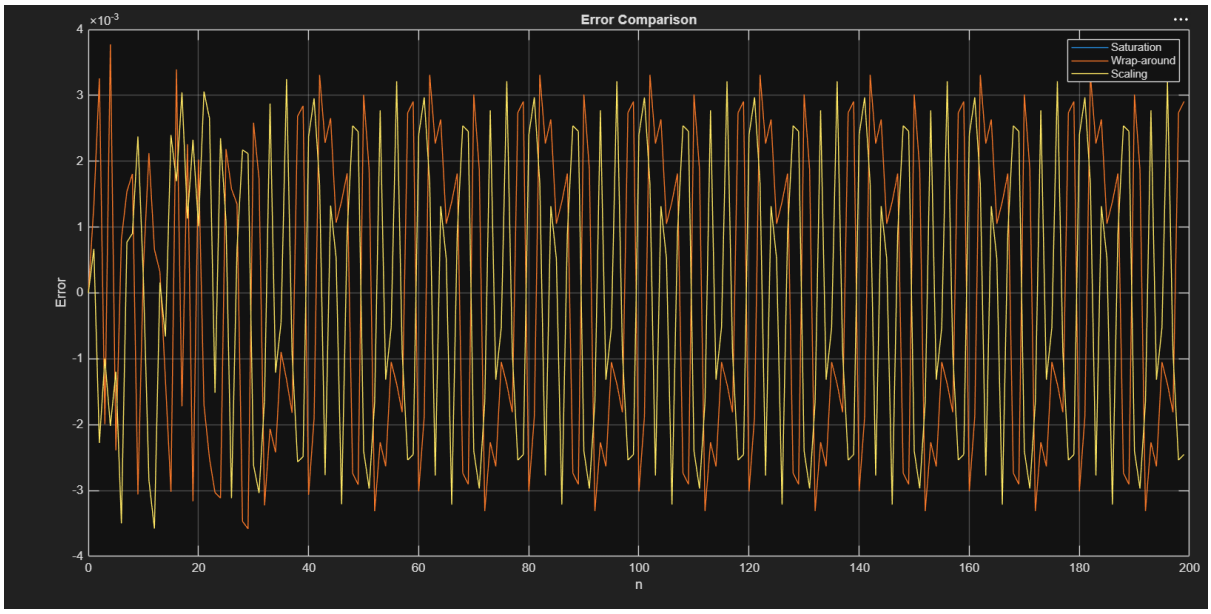


Fig.11.4

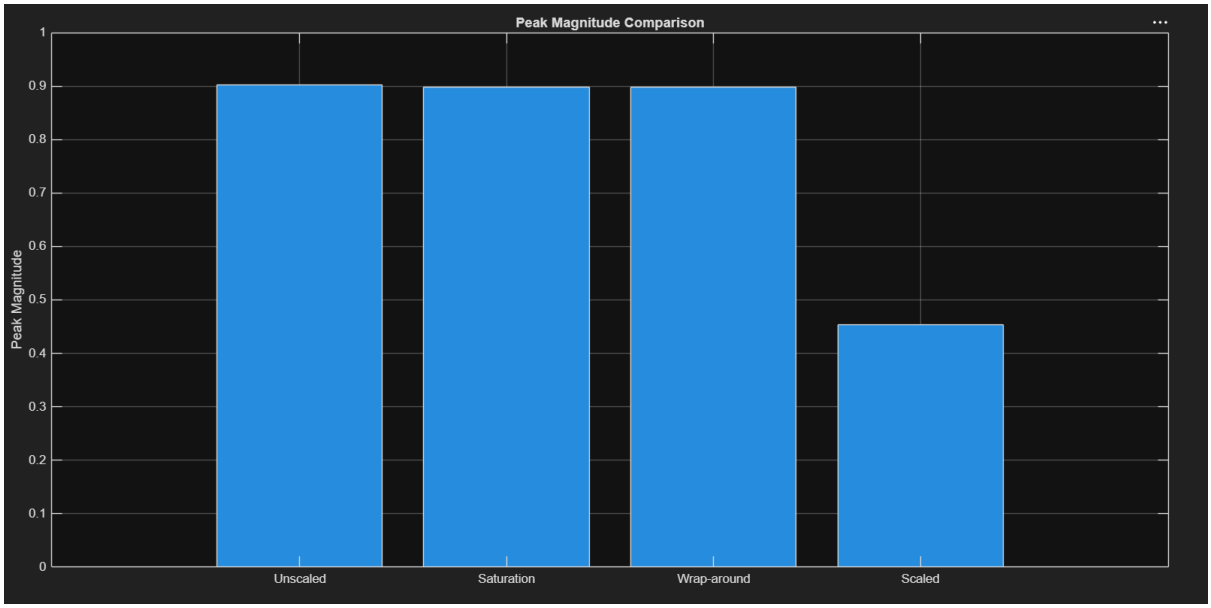

Peak Magnitude Comparison
Peak Magnitude
1
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
Unscaled
Saturation
Wrap-around
Scaled

Fig.11.5

Experiment 12: Implementation and Performance Analysis of Decimation and Interpolation in Multirate Signal Processing Using MATLAB

Output:

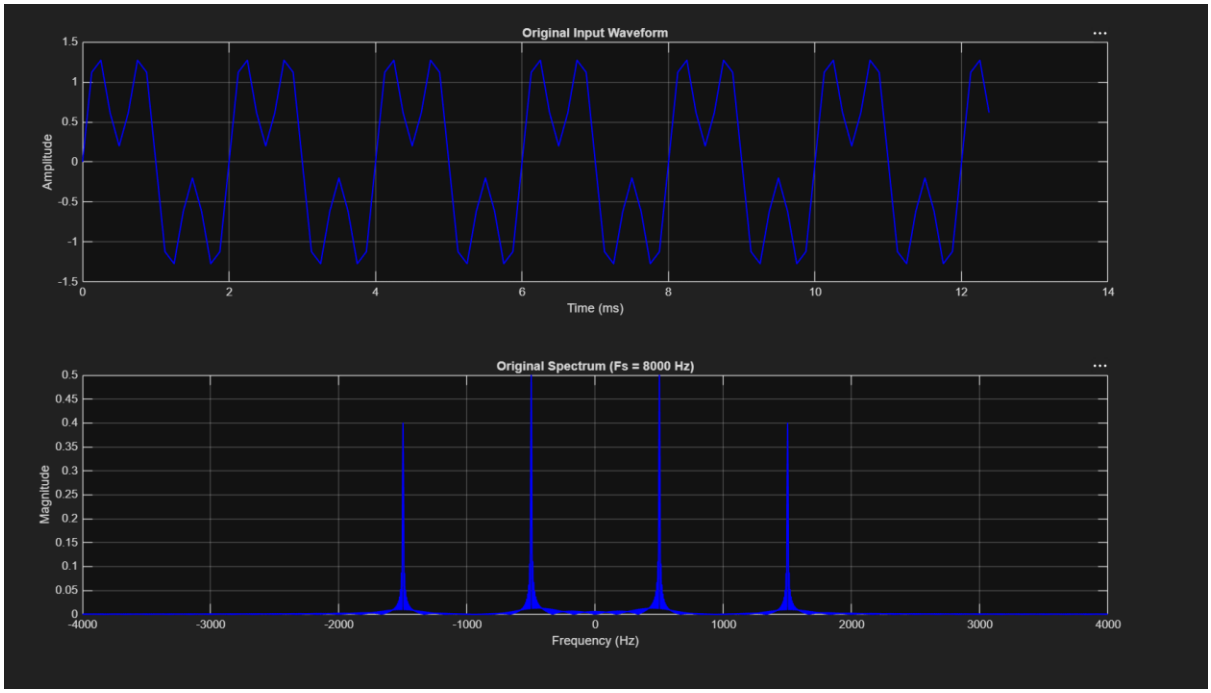


Fig.12.1



Fig.12.2

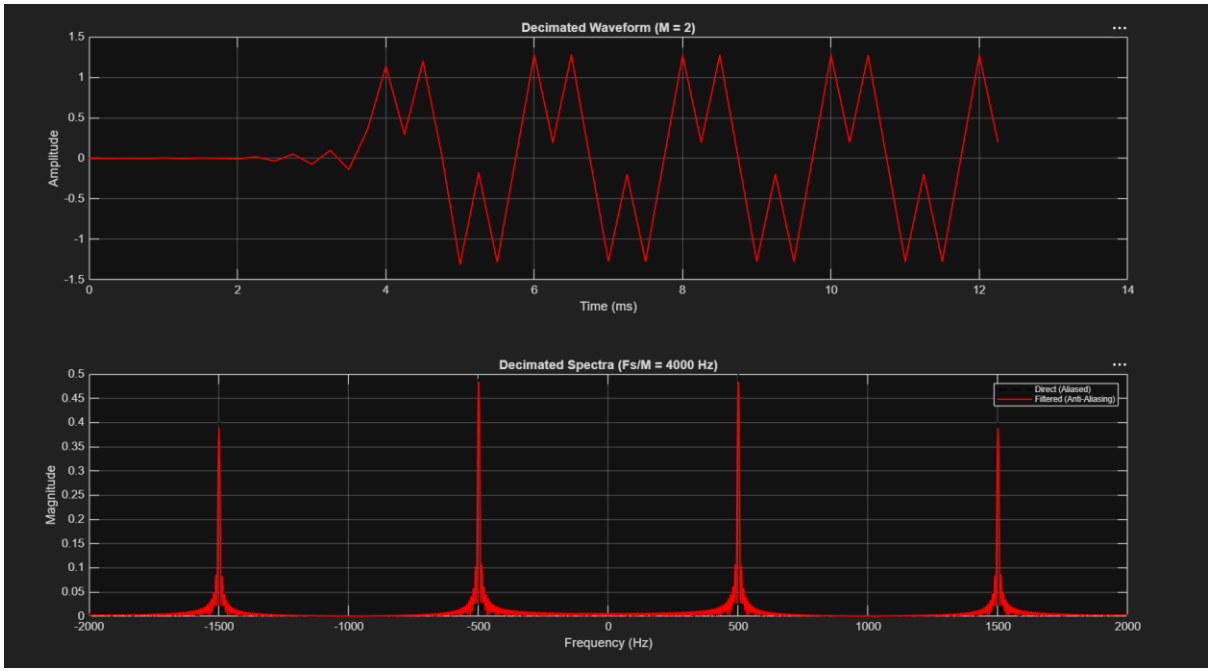

Decimated Waveform (M = 2)
Amplitude
Time (ms)
Decimated Spectra (Fs/M = 4000 Hz)
Magnitude
Frequency (Hz)
Direct (Aliased)
Filtered (Anti-Aliasing)

Fig.12.3

Experiment 13: Polyphase Implementation of Decimation and Interpolation Filters Using MATLAB

Output:

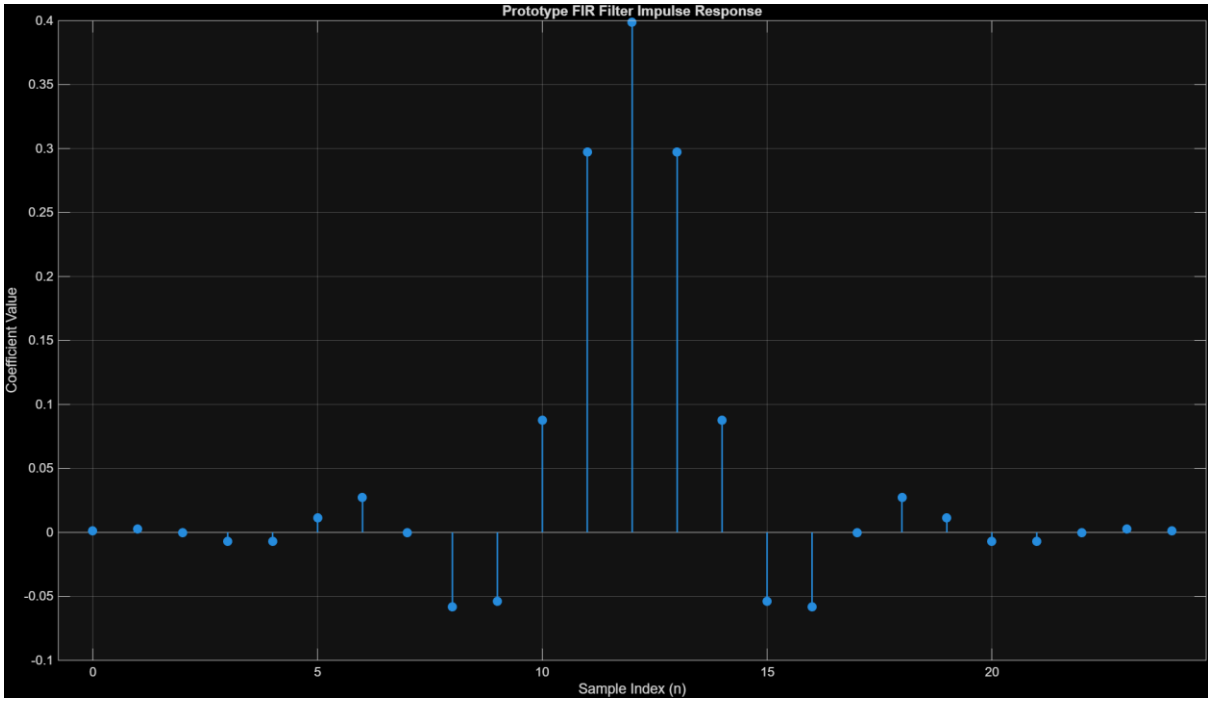


Fig.13.1

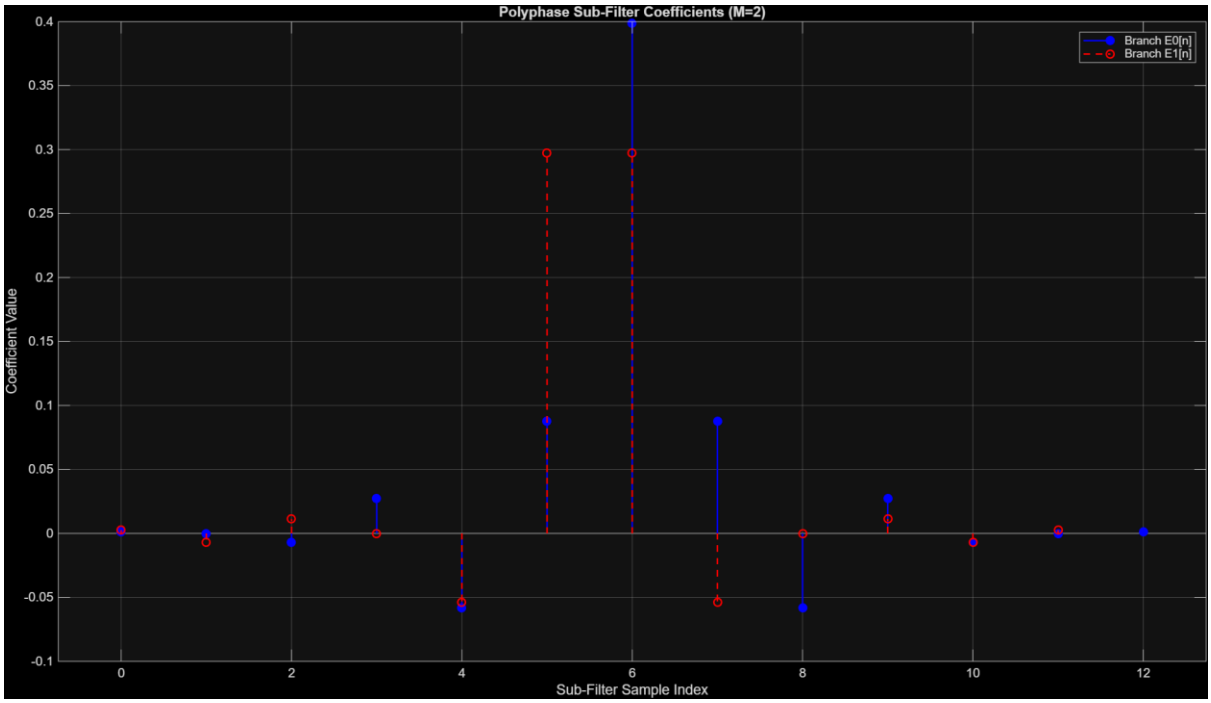


Fig.13.2

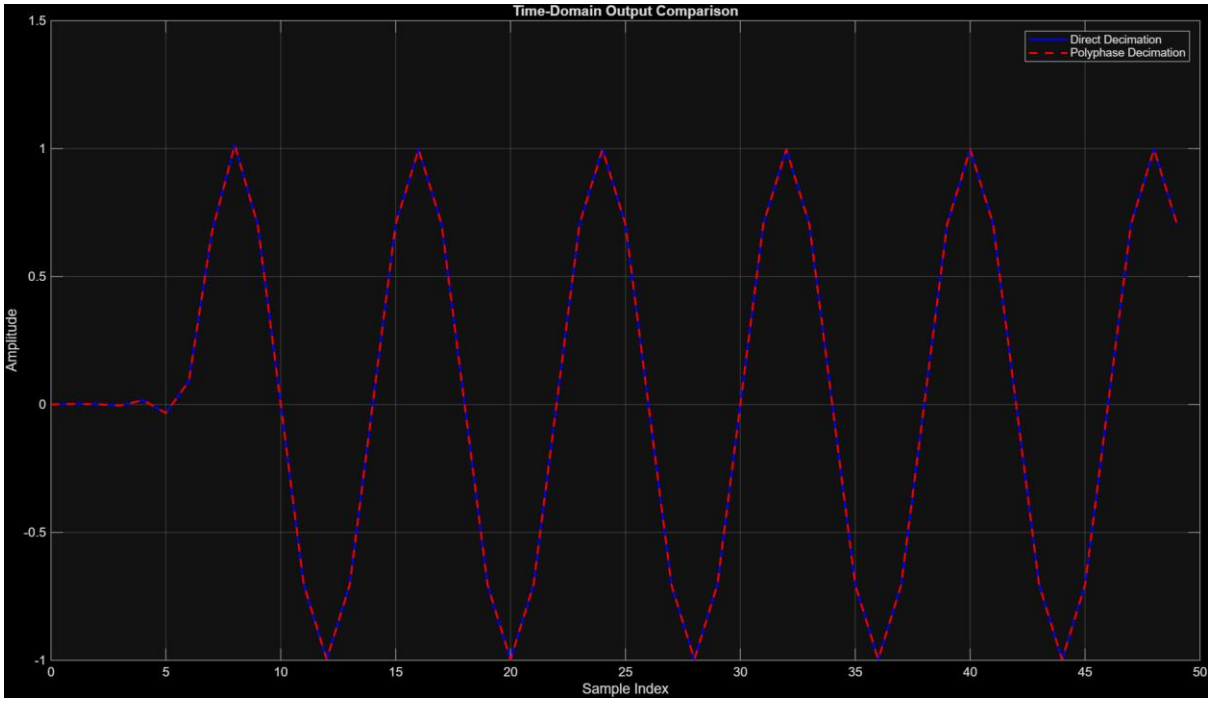
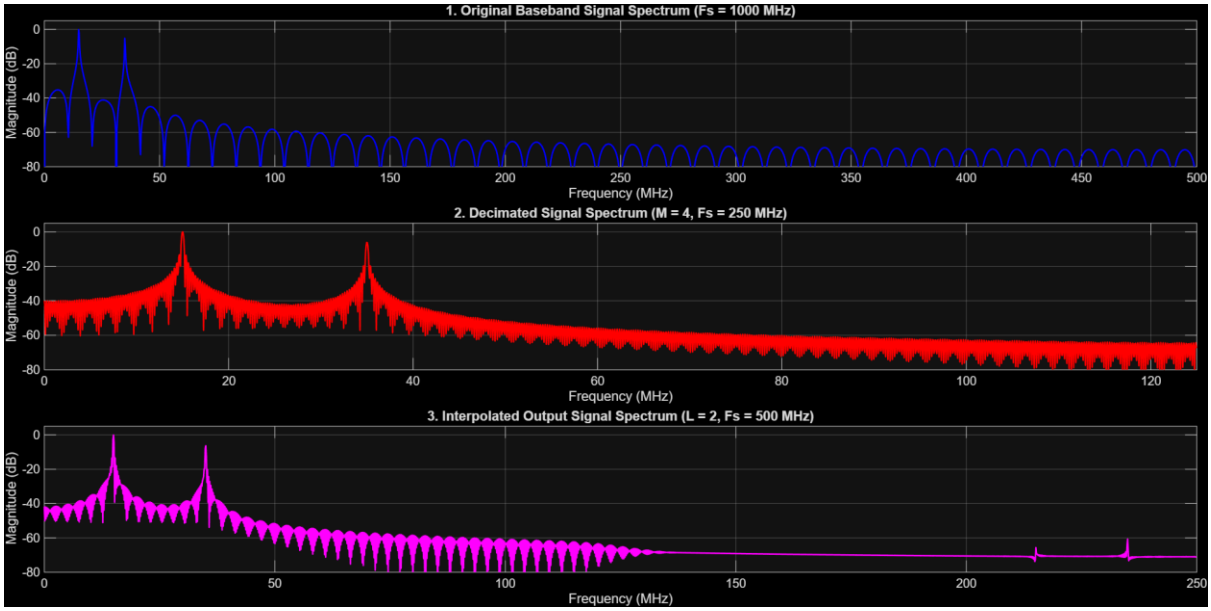

Time-Domain Output Comparison
Direct Decimation
Polyphase Decimation
Amplitude
Sample Index
1.5
1
0.5
0
-0.5
-1
0
5
10
15
20
25
30
35
40
45
50

Fig.13.3

Experiment 14: 5G-Oriented Decimation and Interpolation for Multirate mmWave Signal Processing Using MATLAB

Output:



Experiment 19: Implementation of Spectral Subtraction for Speech Noise Reduction Using MATLAB

Output:

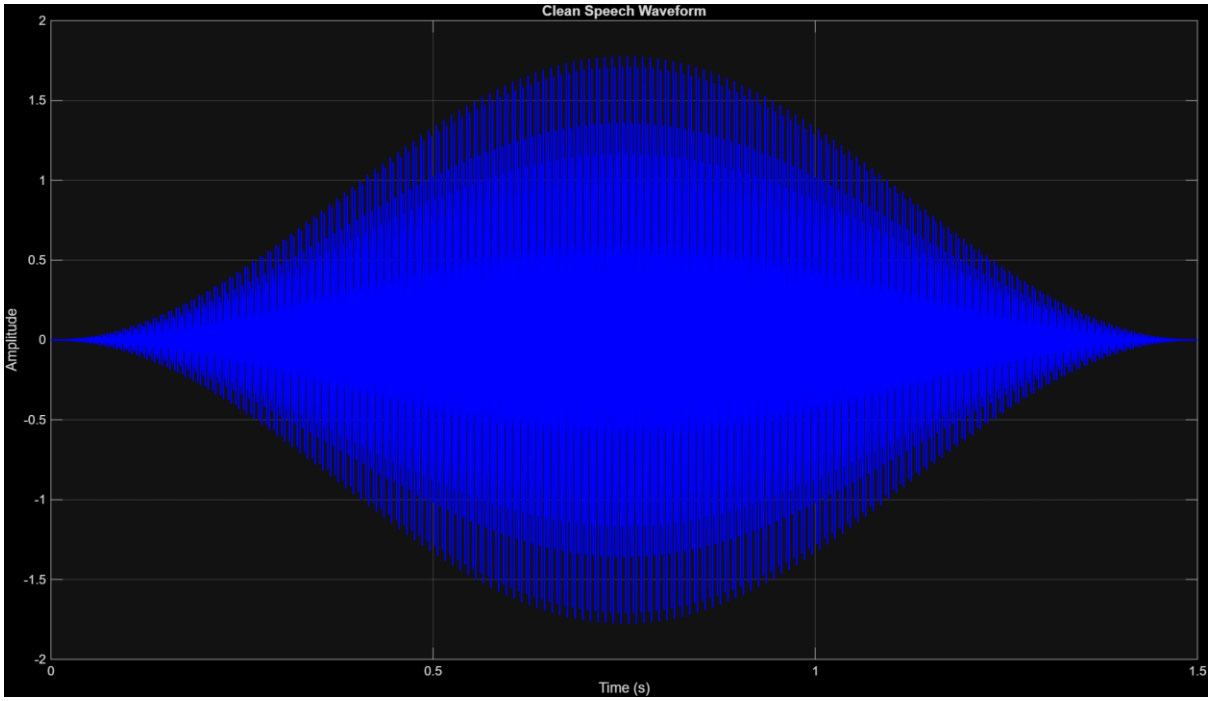


Fig.19.1

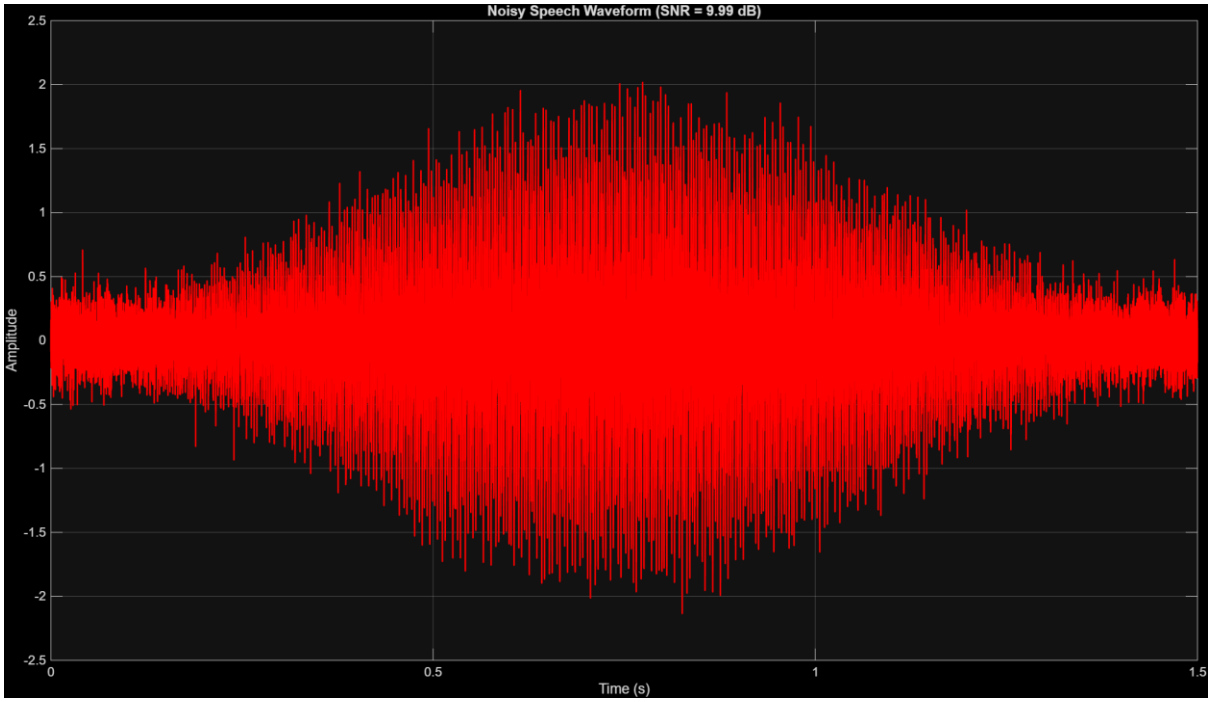


Fig.19.2

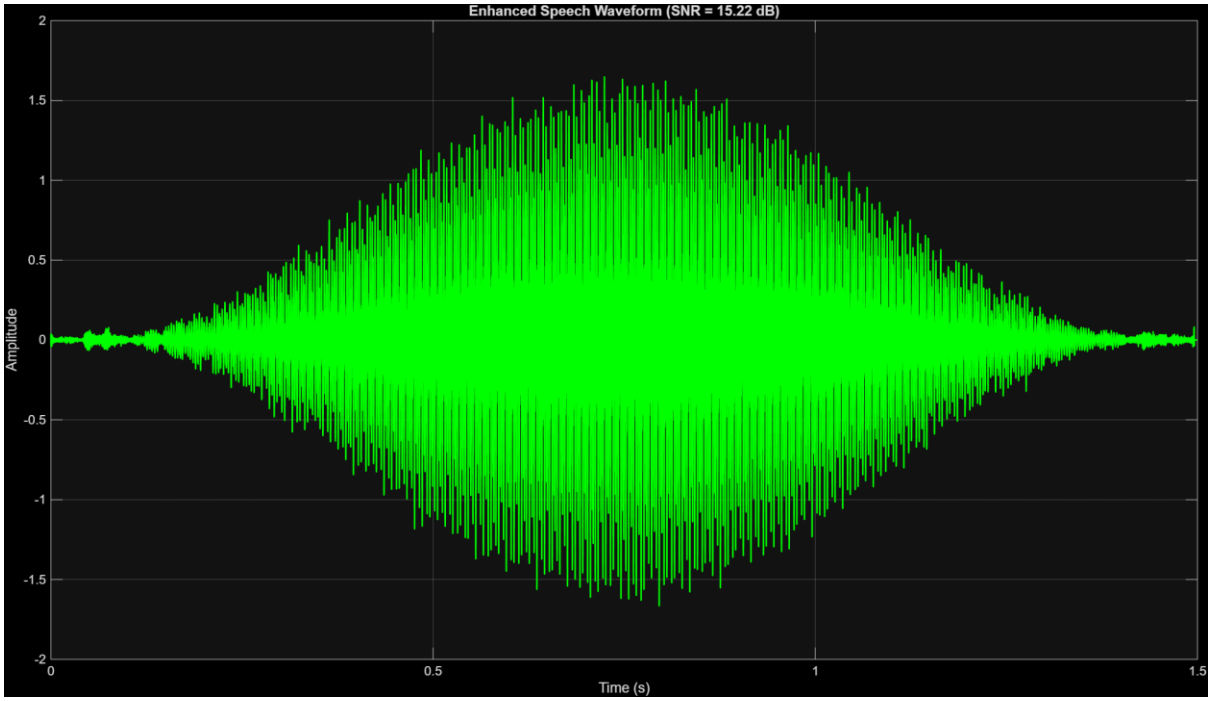

Enhanced Speech Waveform (SNR = 15.22 dB)
Amplitude
Time (s)
2
1.5
1
0.5
0
-0.5
-1
-1.5
-2
0
0.5
1
1.5

Fig.19.3

Experiment 20: LPC-Based Speech Analysis and Vocoder Implementation Using MATLAB

Output:

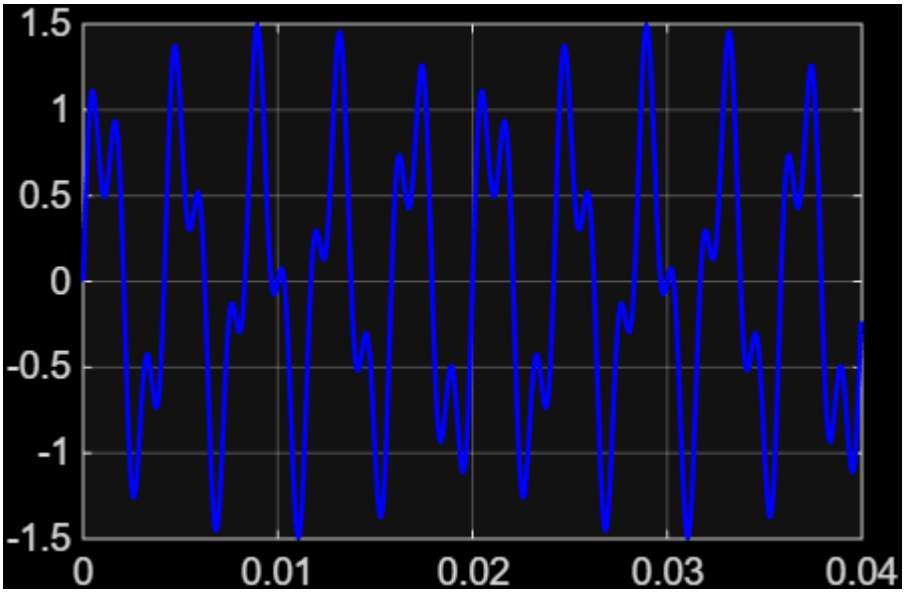


Fig.20.1

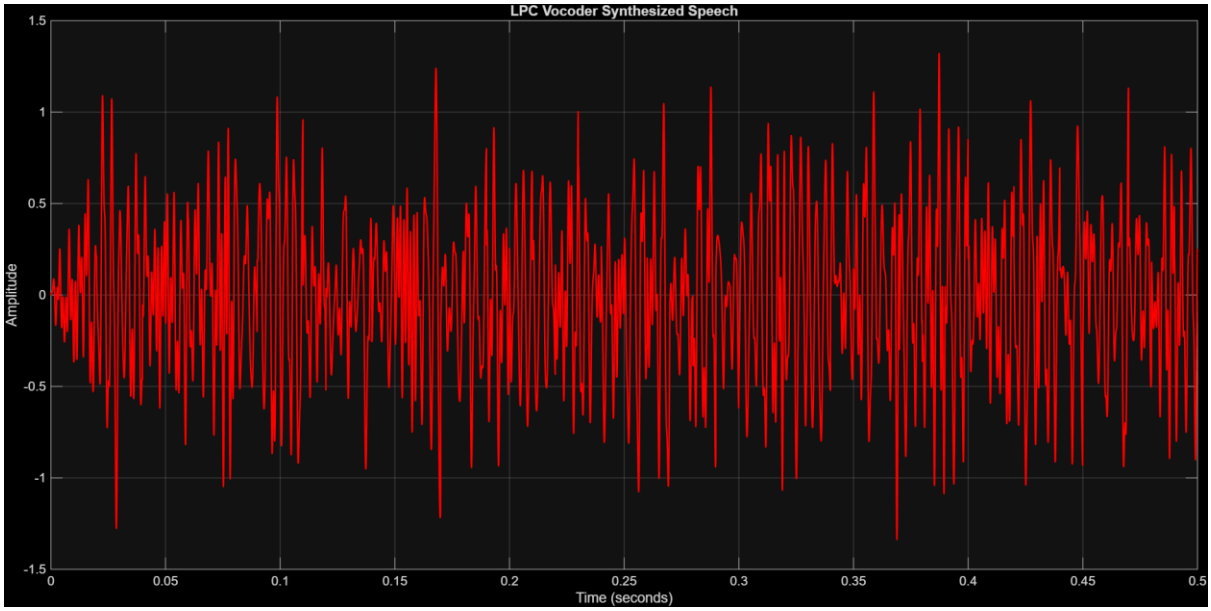


Fig.20.2

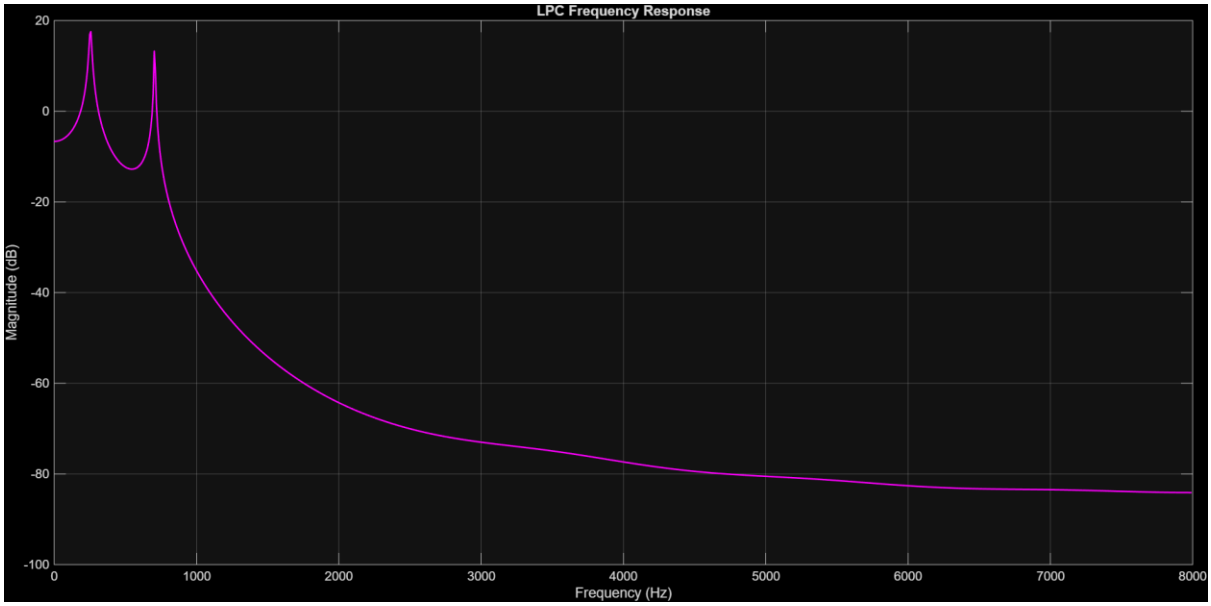


Fig.20.3

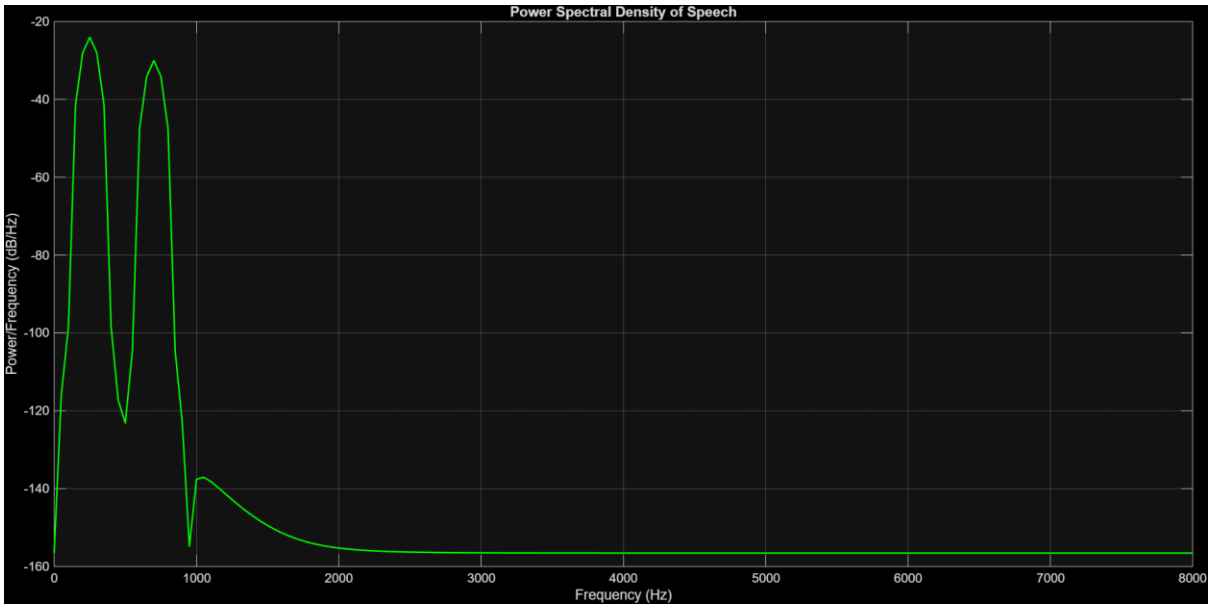


Fig.20.4

Experiment 21: Implementation and Performance Analysis of LMS and NLMS Algorithms for Adaptive Echo Cancellation and Equalization Using MATLAB

Output:

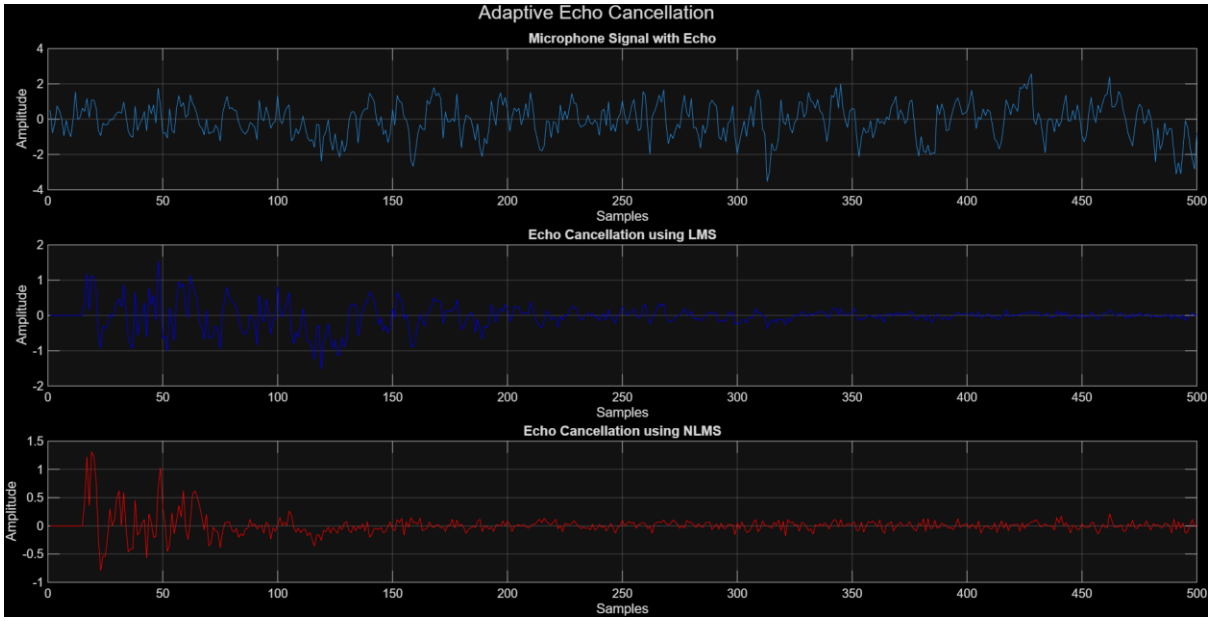


Fig.21.1

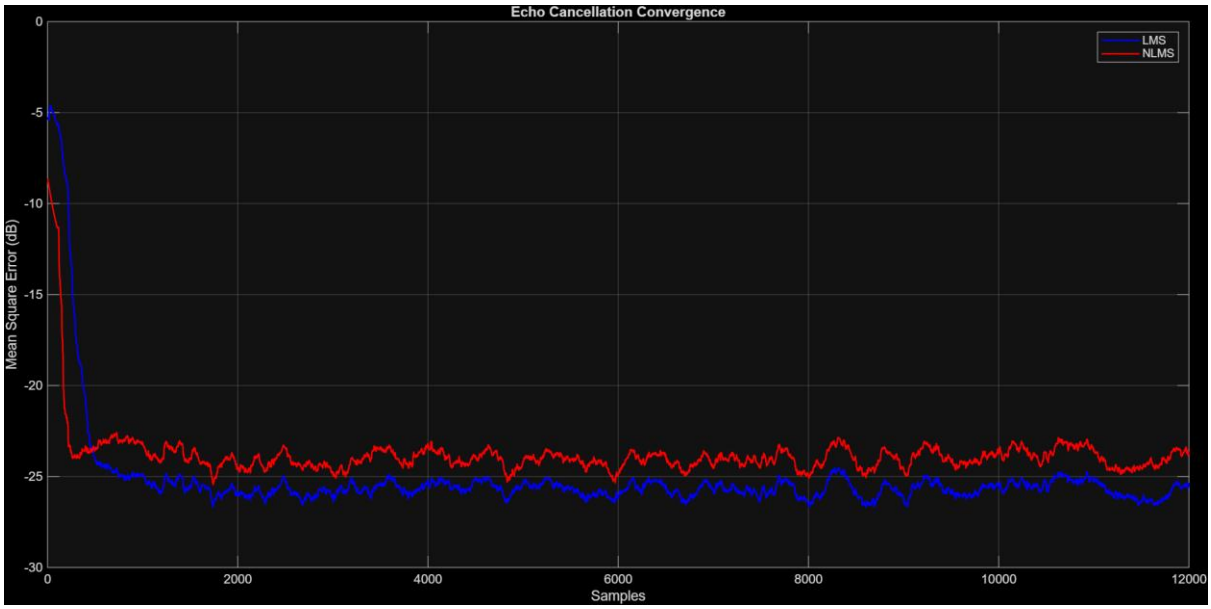


Fig.21.2

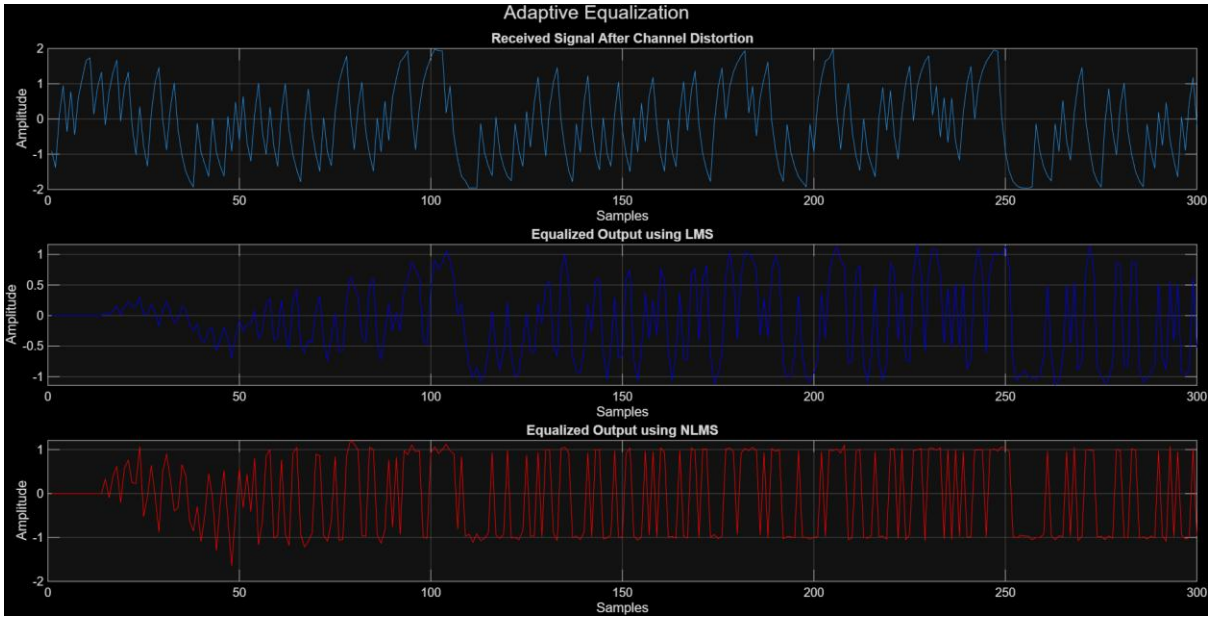


Fig.21.3

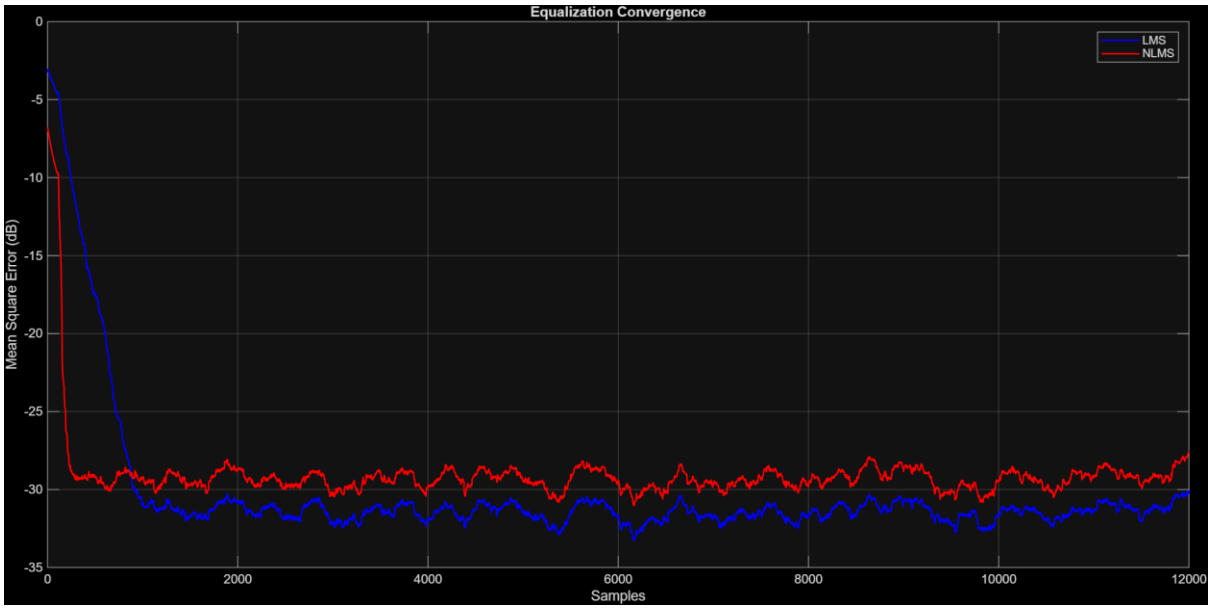


Fig.21.4

Experiment 23: DSP-Based Image Enhancement, Compression and Edge Detection Using MATLAB

Output:



Fig.23.1

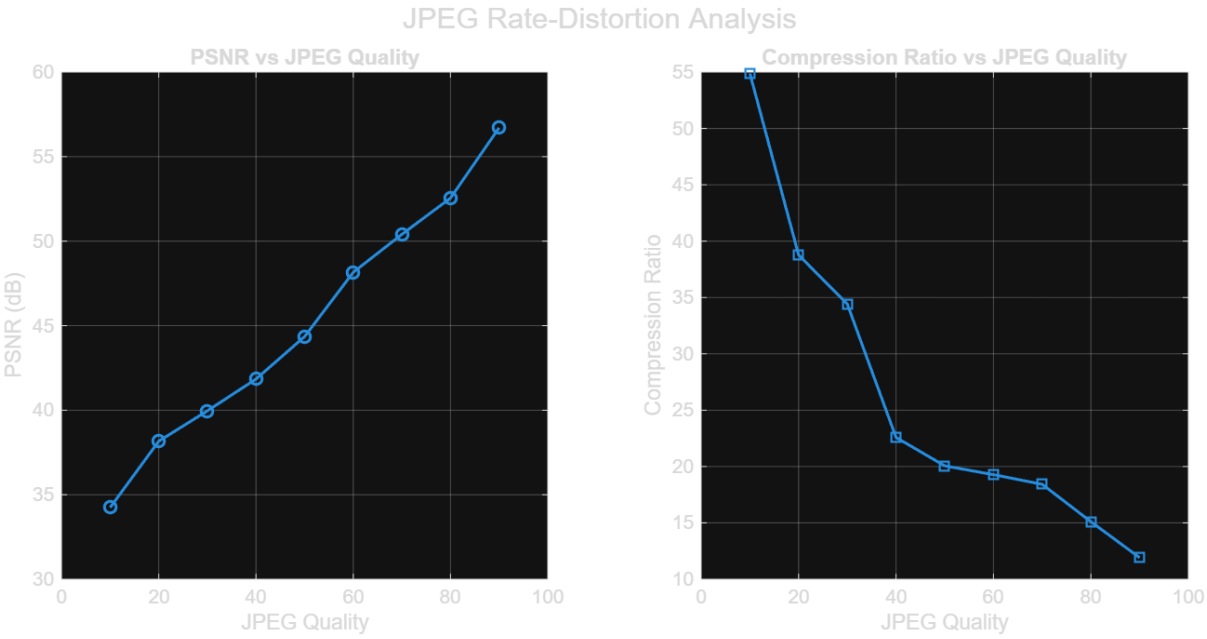


Fig.23.2

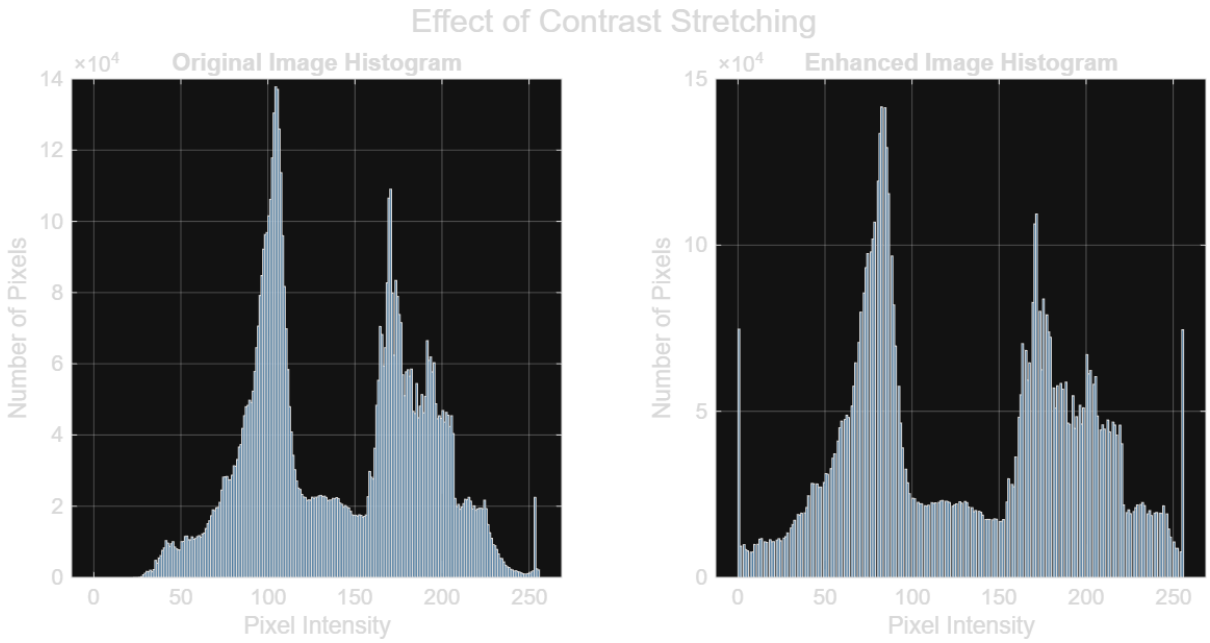


Fig.23.3

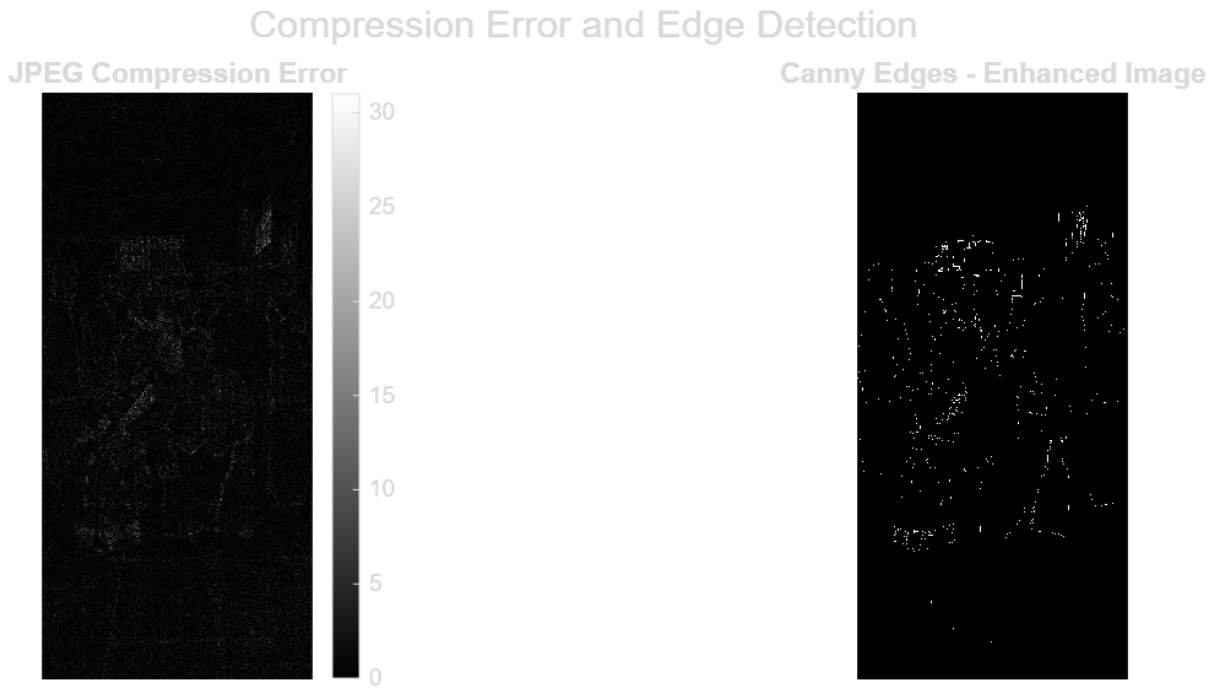


Fig.23.4

Experiment 24: OFDM Channel Estimation Using Least Squares and Minimum Mean Square Error Techniques Using MATLAB

Output:

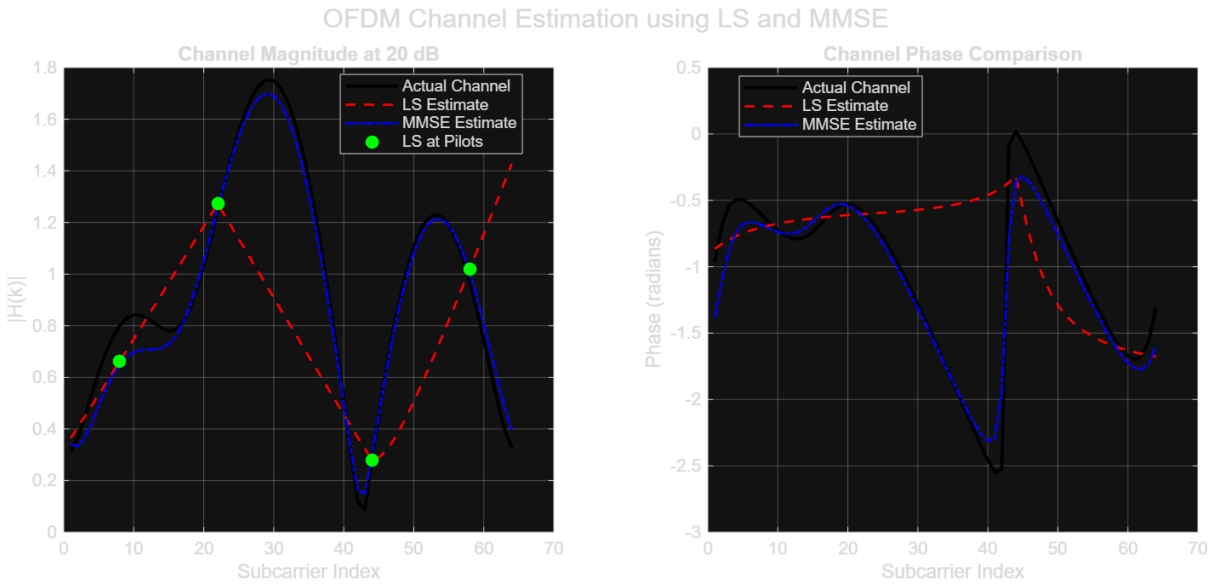


Fig.24.1

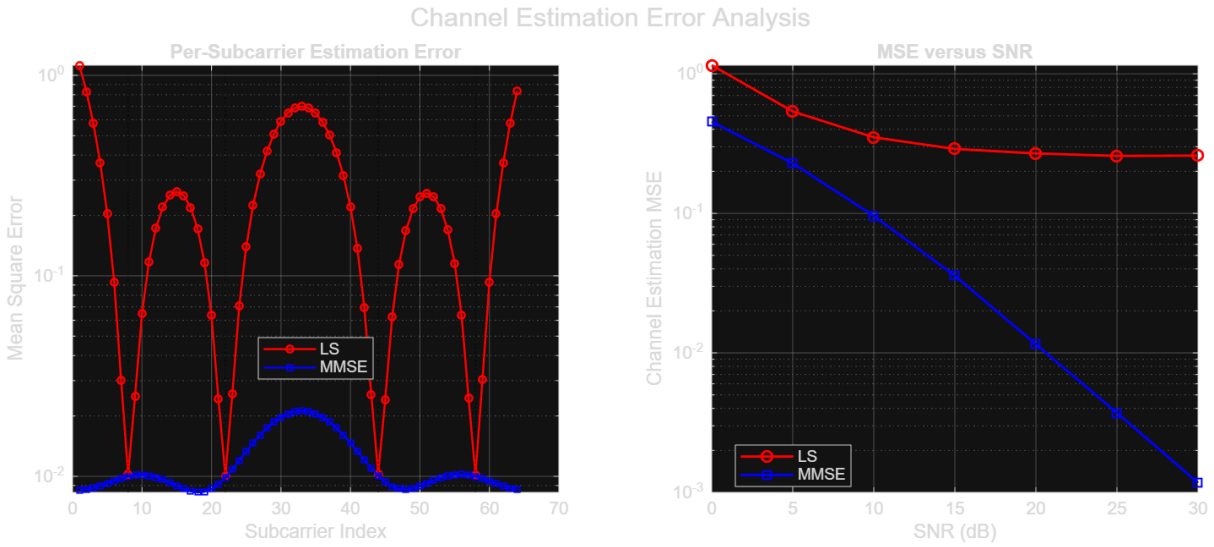


Fig.24.2

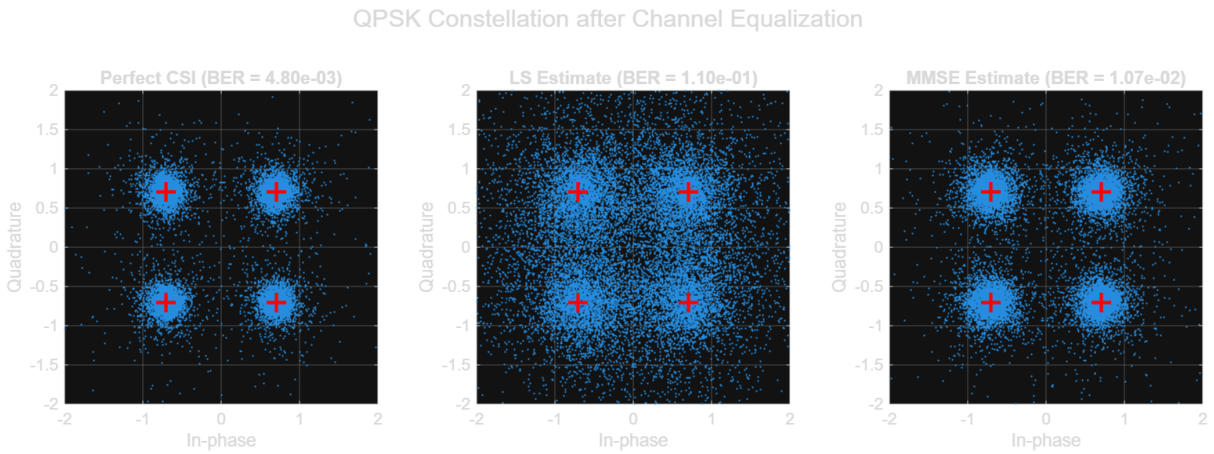


Fig.24.3

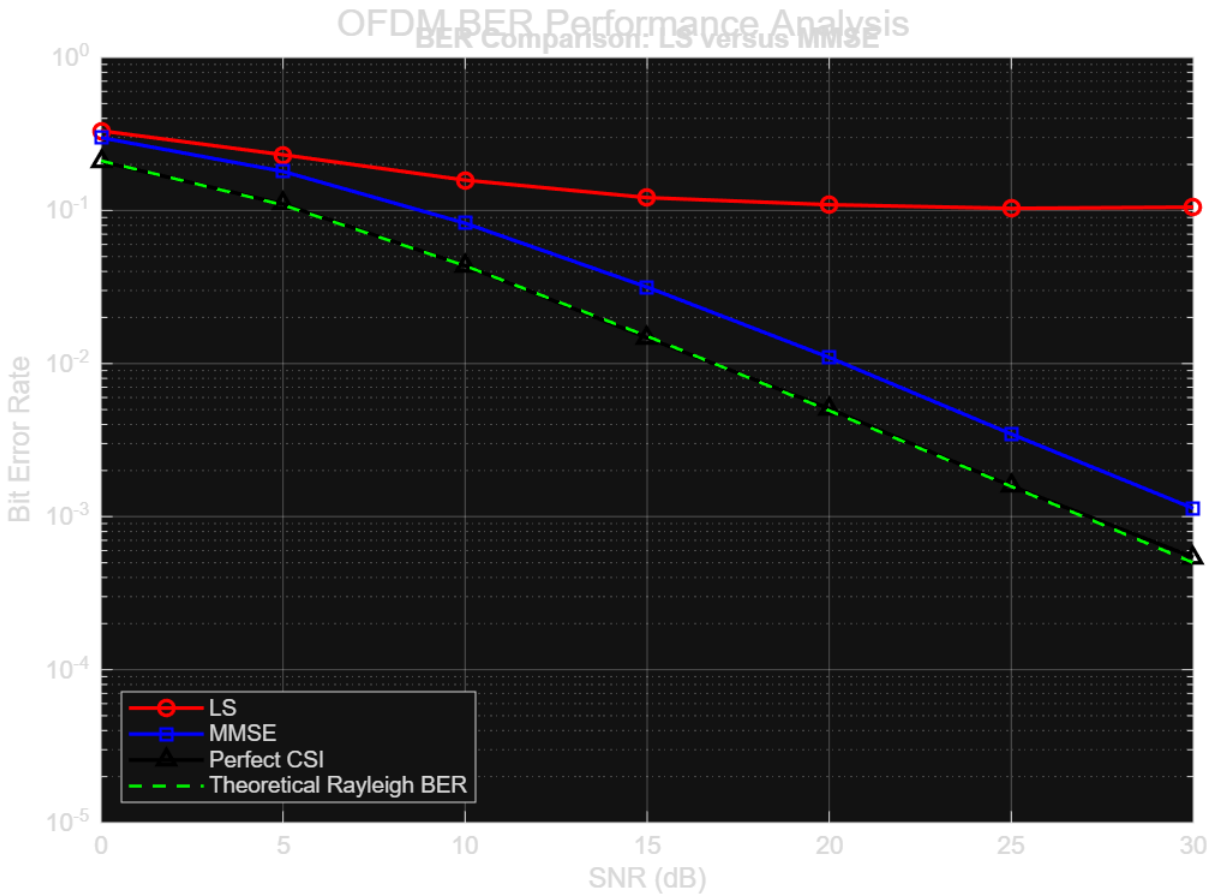


Fig.24.4

Experiment 25: FFT-Based Beamforming for 5G Wireless Communication Using MATLAB

Output:

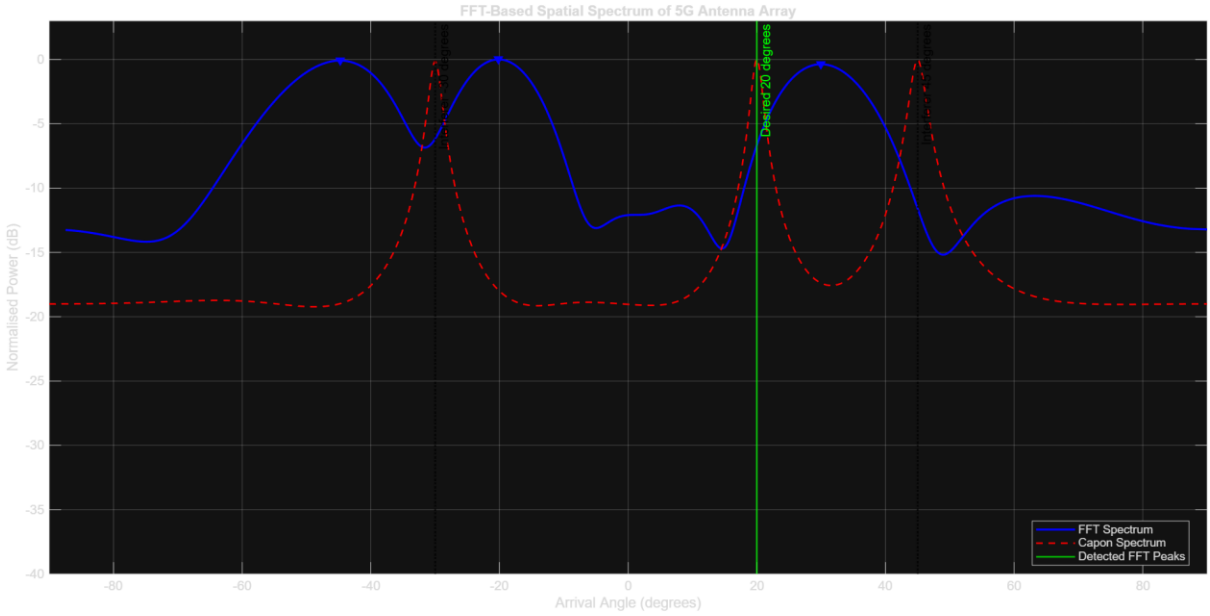
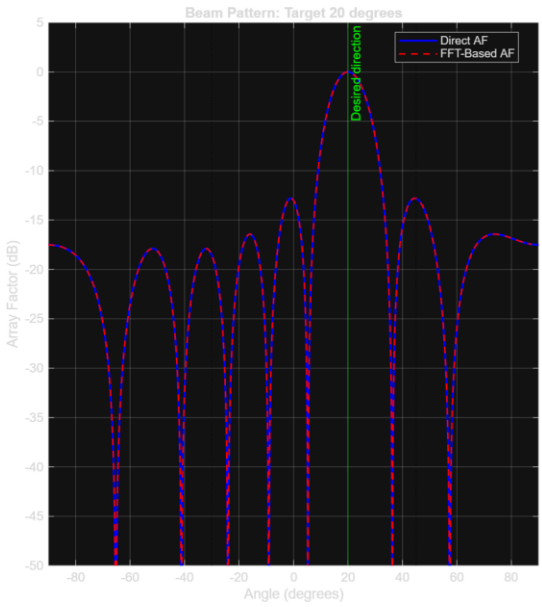


Fig.25.1



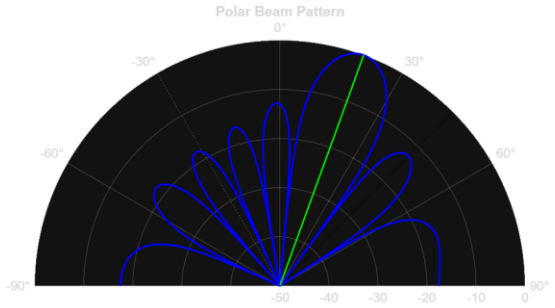


Fig.25.2

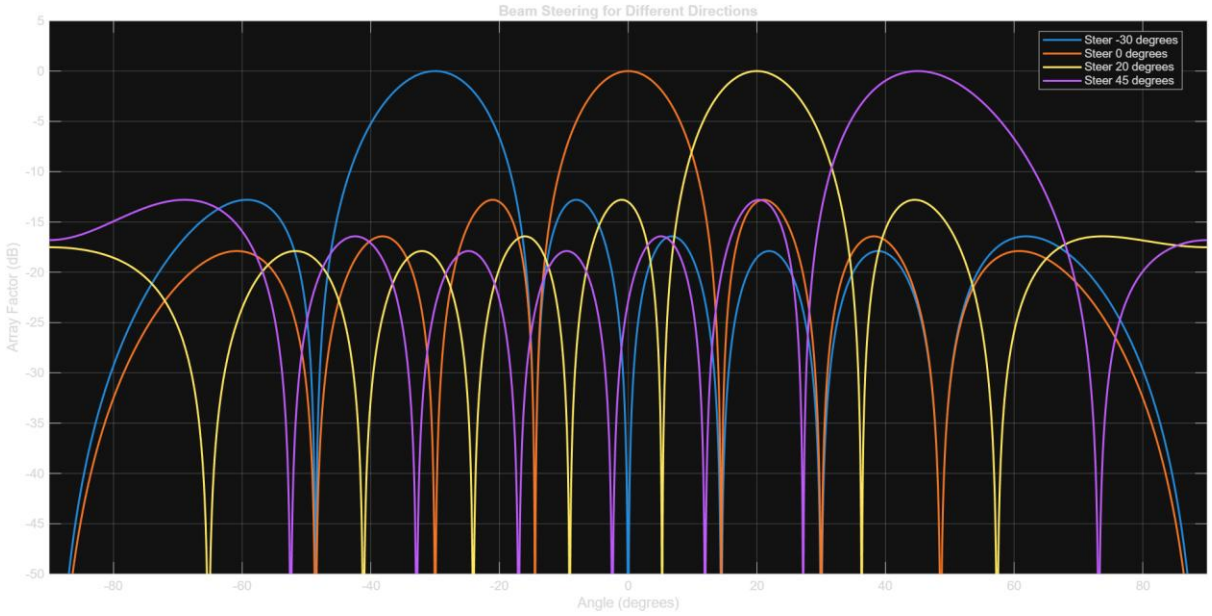


Fig.25.3

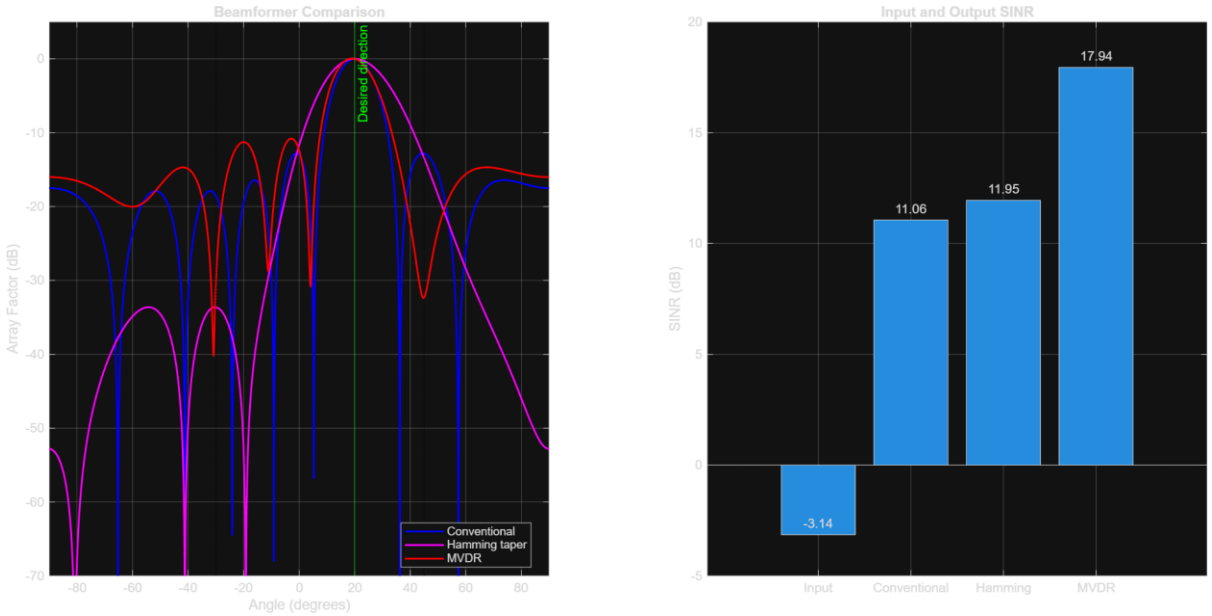


Fig.25.4